

INT422 IT Project Management

Unit III

Agile Project Management

Lecture 16

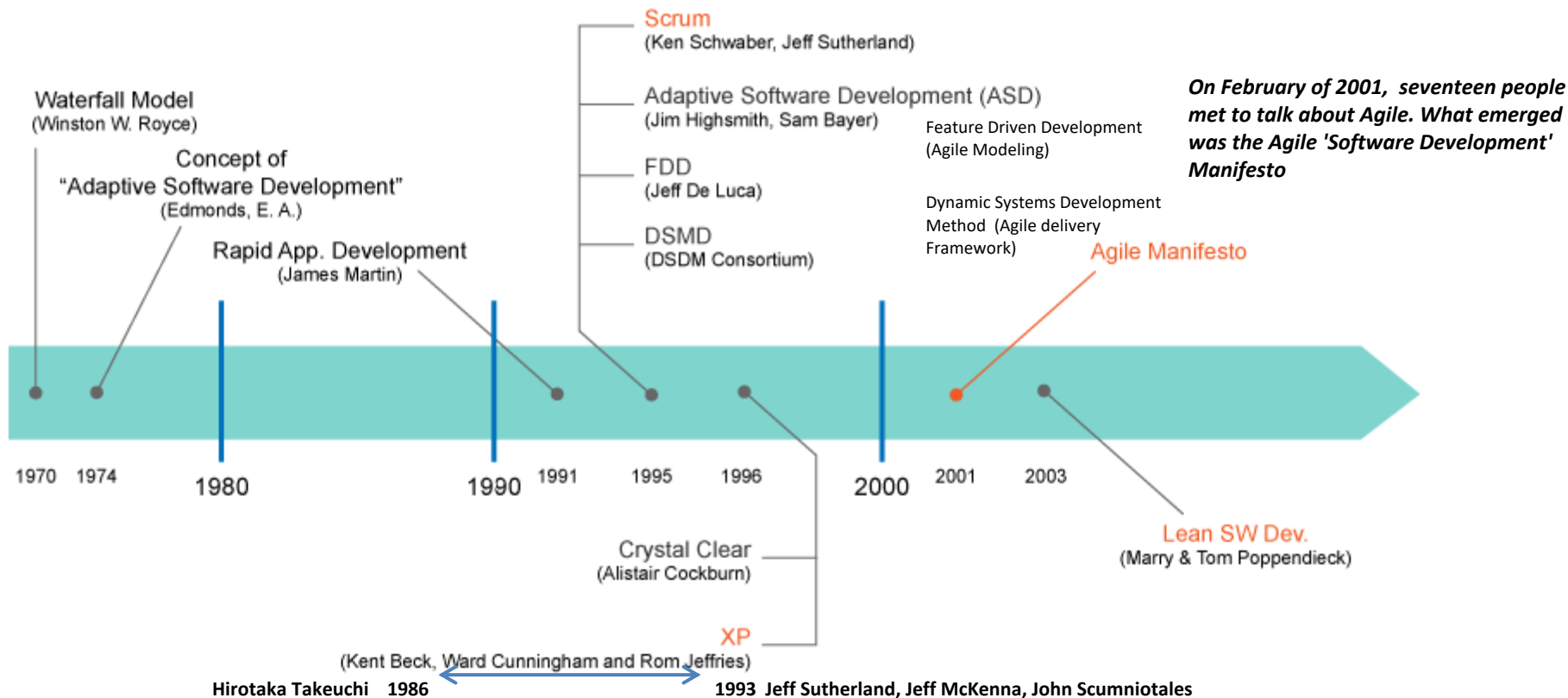
Outline

- Introduction,
- Agile History
- Agile Principles,
- Agile methodologies,
- SAFe: Agile Software Development
- Agile Certified Professional

Agile History

- ❑ In the early 1990s, software development faced a crisis. Industry experts estimated that the time between a validated business need and an actual application in production was about three years
- ❑ Within the space of three years, requirements, systems, and even entire businesses were likely to change. That meant that many projects ended up being cancelled partway through, and many of those that were completed didn't meet all the business's current needs, even if the project's original objectives were met
- ❑ In certain industries, the lag was far greater than three years. In aerospace and defense, it could be 20 or more years before a complex system went into actual use. The Space Shuttle program, which operationally launched in 1982, used information and processing technologies from the 1960s
- ❑ In 1990s, several technology leaders frustrated with these long lead times and decisions made early in a project that couldn't be changed late, began informal talks about ways to develop software more simply, without the process and documentation overhead of **Waterfall** and other popular software engineering techniques of the time

Agile History Timeline



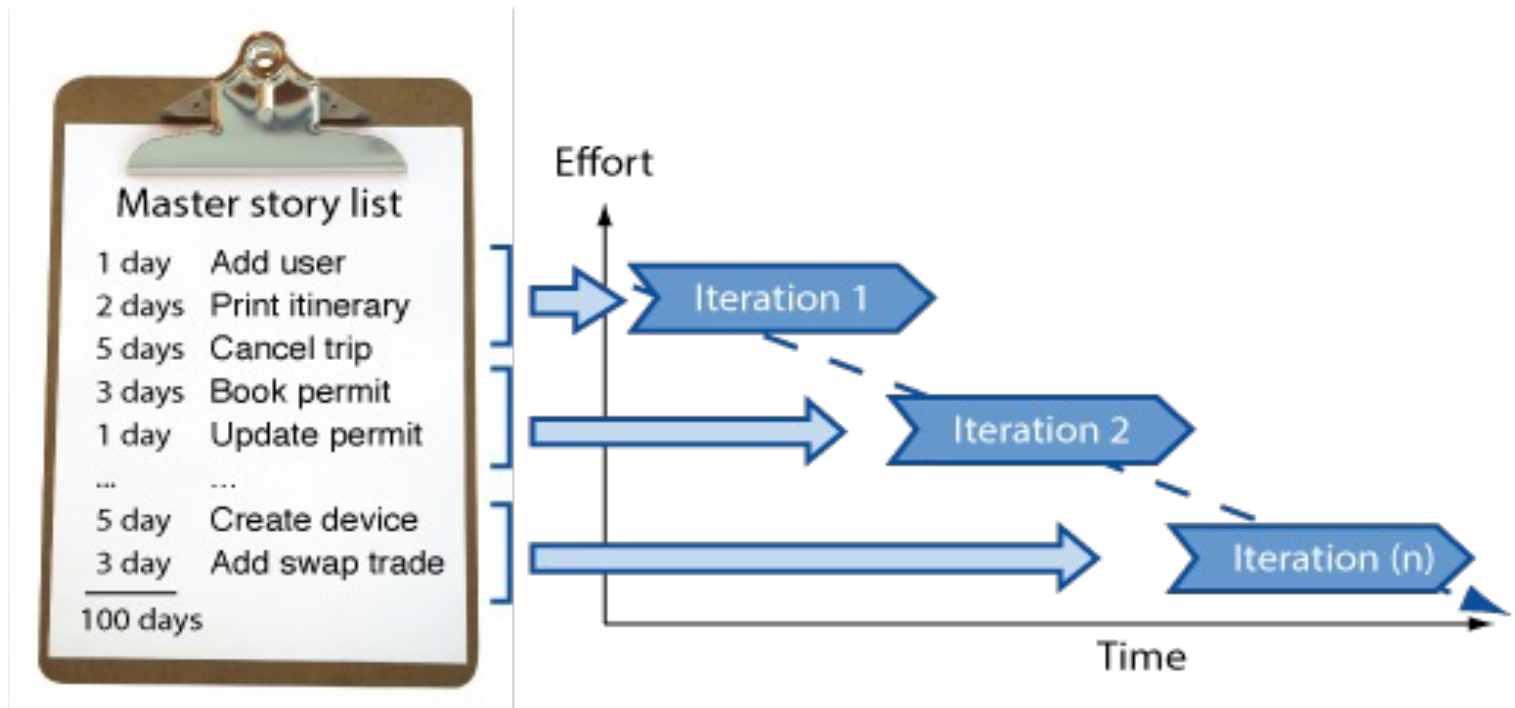
Evolution of Project Management

Simple Comparison	
Traditional Project Management	Modern Project Management (Agile)
Elements impacting execution of a project: a) Planning b) Control	Elements impacting execution of a project: a) Competitive environment b) Creativity and Innovation c) Planning d) Control
Project Success is measured by having: a) Well defined requirements b) Approved budget c) On-Time Delivery	Project Success is measured by: a) Delivery of Business Value (what does customer want) Less important are: a) Having well defined requirements b) Budget constraints c) Set Schedule Today's world has a much higher level of uncertainty which makes it difficult to always start a project with well-defined requirements
Requires a Project Manager	Project Manager Role is distributed among: a) Product Owner b) Scrum Master c) Project Team

Understanding Agile

Agile is a time boxed, iterative approach to software delivery that builds software incrementally from the start of the project, instead of trying to deliver it all at once near the end.

It works by breaking projects down into little bits of user functionality called User Stories, prioritizing them, and then continuously delivering them in short two week cycles called Iteration



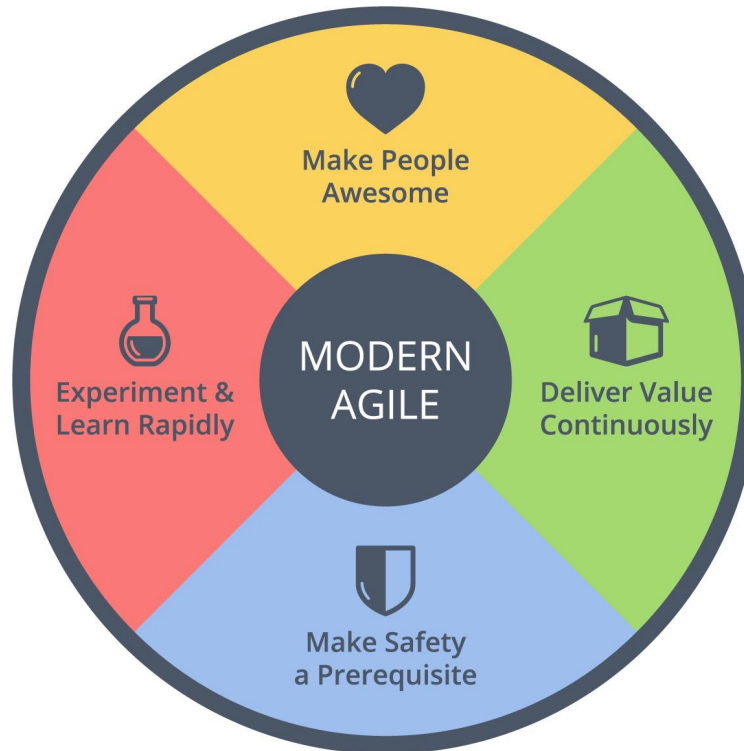
Modern Agile Principles

Experiment & Learn Rapidly

Is a guiding principle of Modern Agile because it protects us from wasting time and helps us discover success faster

Make Safety a Prerequisite

Means establishing safety before engaging in potentially hazardous work



“Make People Awesome”

Amazon has made Customer Obsession a guiding principle since 1997 and it shows. If you make customers awesome, they tend to be natural promoters of your products or services

Deliver Value Continuously

Anything valuable that hasn't been delivered isn't helping anyone. How might we deliver the right outcomes faster

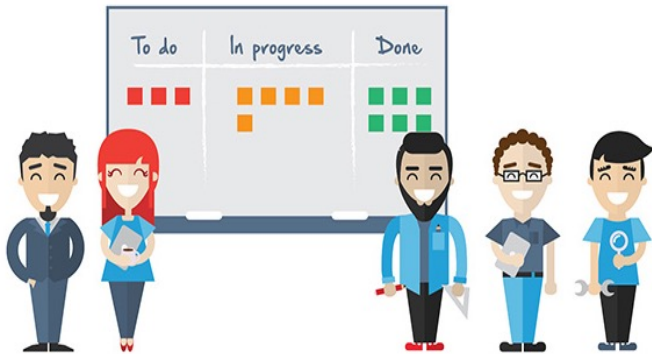
Agile Principles

- #1 Satisfy Customers Through Early & Continuous Delivery**
- #2 Welcome Changing Requirements Even Late in the Project**
- #3 Deliver Value Frequently**
- #4 Break the Silos of Your Project**
- #5 Build Projects Around Motivated Individuals**
- #6 The Most Effective Way of Communication is Face-to-face**
- #7 Working Software is the Primary Measure of Progress**
- #8 Maintain a Sustainable Working Pace**
- #9 Continuous Excellence Enhances Agility**
- #10 Simplicity is Essential**
- #11 Self-organizing Teams Generate Most Value**
- #12 Regularly Reflect and Adjust Your Way of Work to Boost Effectiveness**

Real world example of how to apply modern Agile guiding principles

- Let's say you decide you want to write a book. If you are driven by Modern Agile principles, you'll want to quickly discover if the book will make people awesome.
- It would not be a safe investment of your time if no one has interest in your book. So, to learn whether there is genuine excitement about your topic, you'll need to experiment and learn rapidly.
- To do that, you'll need to deliver some value to some people and learn rapidly from their response. Perhaps you write a few pages, a short chapter or a small article, receiving feedback from an inner circle and then from a wider audience.
- As you write and learn, you'll want to make sure your work is safe by using tools that protect your work by backing it up.
- If you are interviewing someone important for the book, you may want multiple recorders running, so if one breaks or runs out of space, you'll have built-in fault tolerance to ensure you get a good recording.

Scrum, Kanban & XP Agile Methodologies



Kanban (an industrial engineer at Toyota, developed Kanban to improve manufacturing efficiency) is also a tool used to organize work for the sake of efficiency. Where Scrum limits the amount of time allowed to accomplish a particular amount of work (by means of sprints), Kanban limits the amount of work allowed in any one condition (only so many tasks can be ongoing, only so many can be on the to-do list.)

Scrum is a tool used to organize work into small, manageable pieces that can be completed by a cross-functional team within a prescribed time period (called a sprint, generally 2-4 weeks long) to plan, organize, administer, and optimize this process.

Extreme Programming (XP) is an agile software development framework that aims to produce higher quality software, while focusing on customer satisfaction by delivering what's needed when needed. Its guiding principles are: Communication, Simplicity, Feedback, Respect and Courage

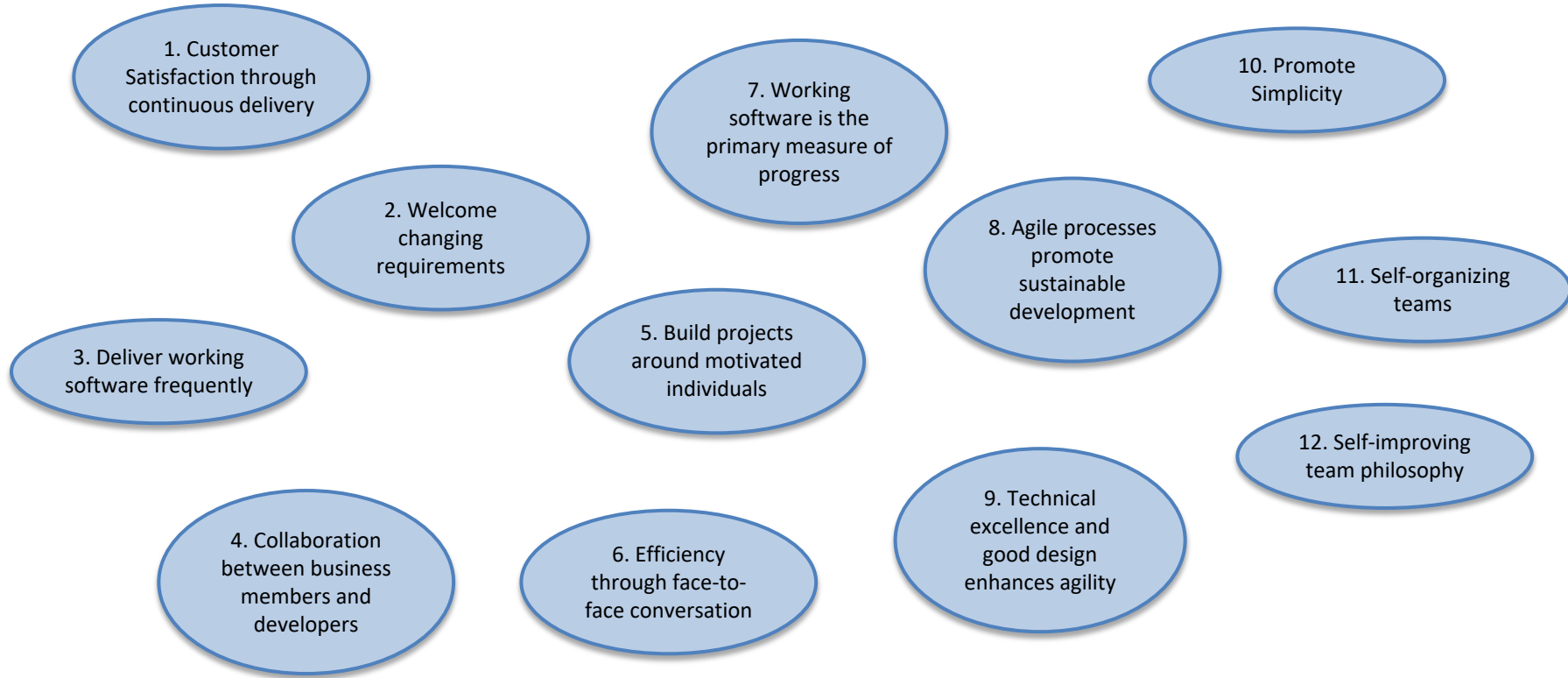


Relationship Between Agile Scrum, Lean

- ❑ Scrum teams typically work in iterations (called sprints) that are from two weeks to one month long. XP teams typically work in iterations that are one or two weeks long
- ❑ Scrum teams do not allow changes into their sprints. XP teams welcome changes as long as work has not started
- ❑ Extreme Programming teams work in a strict priority order. Scrum teams have flexibility to choose what prioritized features to work on
- ❑ Scrum doesn't prescribe any engineering practices. XP does (things like test-driven development, focus on automated testing, pair programming, simple design, refactoring, and so on)

	Lean	Scrum
Goal	Reducing waste (process)	User-centered product (adapting to needs)
Timeline	Continuous improvement	Sprint-driven (1–4 weeks at a time)
Outcome	Improving process output/reducing costs	Working product increment (physical deliverable)
Origins	Industrial manufacturing	Software development
Scale	Company-wide	Often implemented on a team basis
Management	Top-down but more respect for worker opinions	Self-organization (if a scrum master exists, it is a facilitator)
Applied to	Existing repeatable processes	Developing new solutions

Agile Manifesto

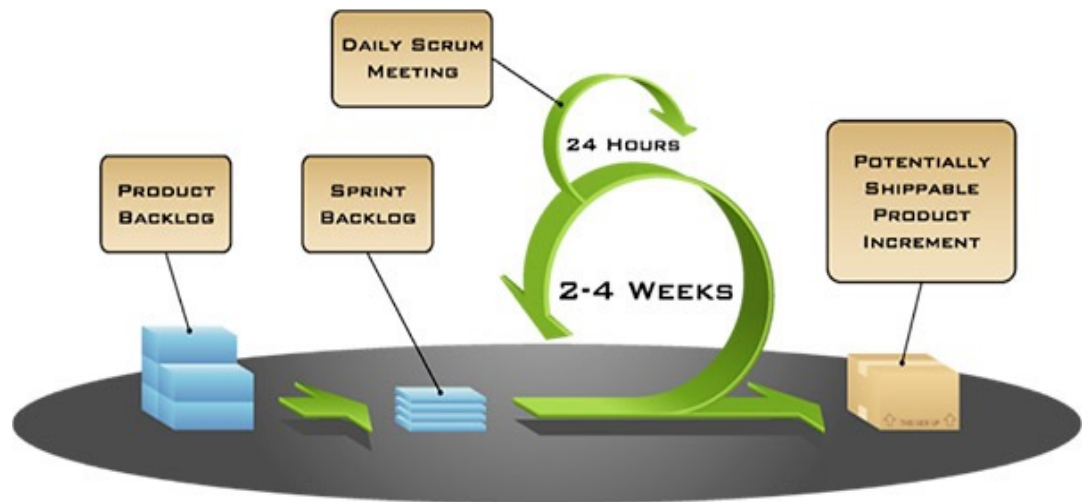
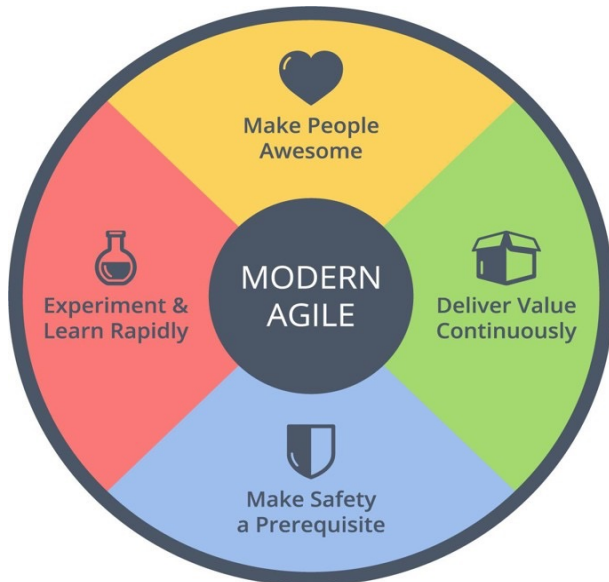


The Four Values of The Agile Manifesto

- Individuals and Interactions Over Processes and Tools
- Working Software Over Comprehensive Documentation
- Customer Collaboration Over Contract Negotiation
- Responding to change over following a Plan

What is Modern Agile?

- The industry has become overloaded with techniques, processes, methodologies and tools that have fallen under the Agile umbrella.
- Many consider these more marketing hype than Agile, feeling they are moving away from the simplicity that Agile principles promote.
- Modern Agile aims to move Agile to the next level, not by adding complexity, but by simplifying— stressing only adherence to four principles



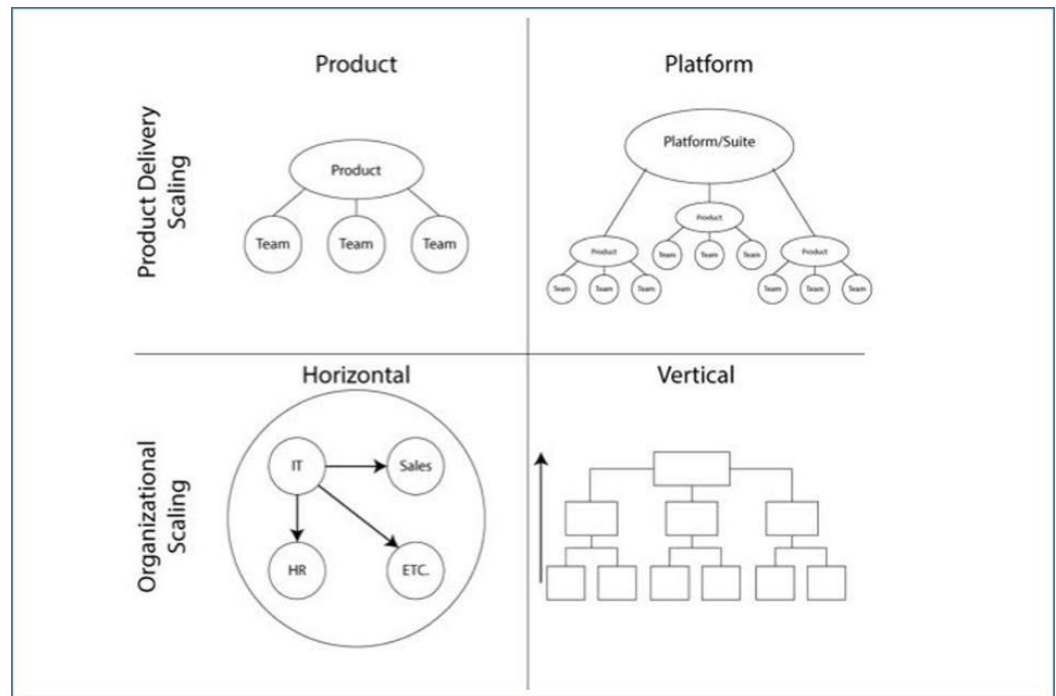
What is Scaling Framework?

- Scaling is the application of practices of “Agility” across “Horizontal” and “Vertical” space through an organization. Scrum (a tool used to organize work into small, manageable pieces that can be completed by a cross-functional team within a prescribed time period) is a process unique to software development and can not be applied outside of that paradigm.
- Agile mindset and Lean thinking however can be practiced across all organizational teams.

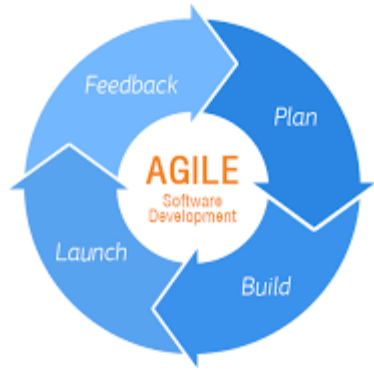
Scrum →

How do we make the entire organization use Agile in their day to day processes? “the Agile Mindset”

Agility →



SAFe: Agile Software Development



Agile software development refers to a group of software development methodologies (*Scrum, Kanban, and Extreme Programming*) based on iterative development, where requirements and solutions evolve through collaboration between self-organizing cross-functional teams

Agile development refers to any development process that is aligned with the concepts of the Agile Manifesto.

The Manifesto was developed by a group leading figures in the software industry, and reflects their experience of what approaches do and do not work for software development

SAFe: Lean Software Development Principles

Lean development can be summarized by seven principles, very close in concept to lean manufacturing principles:

- ☐ Eliminate Waste
- ☐ Amplify Learning
- ☐ Decide as late as possible
- ☐ Deliver as fast as possible
- ☐ Empower the team
- ☐ Build integrity in
- ☐ See the whole

What is SAFe?

The **Scaled Agile Framework (SAFe)**, is intended to guide enterprises in scaling lean and agile practices. SAFe promotes alignment, collaboration, and delivery **across large numbers of agile teams** (Agile on steroids).

It was developed by and for practitioners, by leveraging three primary bodies of knowledge:

1. Agile software development (Agile Software Development is an umbrella term for a set of methods and practices based on the values and principles expressed in the Agile Manifesto. Solutions evolve through collaboration between self-organizing, cross-functional teams utilizing the appropriate practices for their context)
2. Lean product development (Eliminate Waste, Amplify Learning, Decide as late as possible, Deliver as fast as possible, Empower the team, Build integrity, See the whole)
3. Systems thinking (Is a management discipline that concerns an understanding of a system by examining the linkages and interactions between the components that comprise the entirety of that defined system)

Agile Release Train (ART)

- **What is it?**

The ART metaphor describes the program level teams, roles, and activities that incrementally deliver a continuous flow of value. ARTs are virtual organizations formed to span functional boundaries, eliminate unnecessary handoffs and steps, and accelerate value delivery by implementing SAFe Lean-Agile principles and practices

- **Who is responsible for it?**

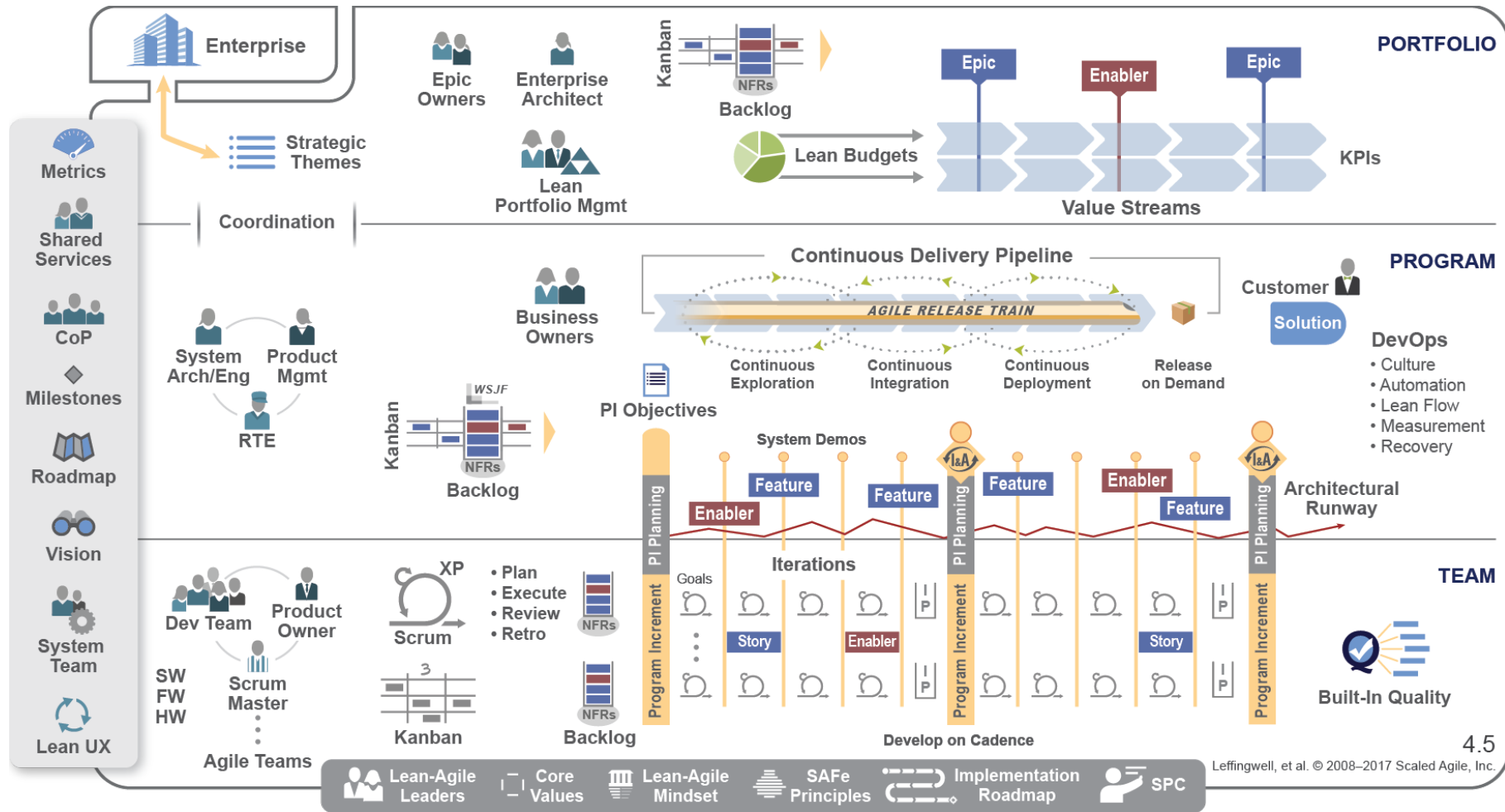
System Architect/Engineer– Is an individual or small cross-discipline team that truly applies Systems Thinking. They define the overall architecture for the system, help define Nonfunctional Requirements (NFRs), determine the major elements and subsystems, and help design the interfaces and collaborations among them.

Product Management – Is the internal voice of the Customer and works with customers and Product Owners to understand and communicate their needs, define system features, and participate in validation. They are responsible for the Program Backlog.

Release Train Engineer (RTE) – Is a servant leader and the chief Scrum Master for the train. The RTE facilitates optimizing the flow of value through the program using various mechanisms, such as the Program Kanban, Inspect & Adapt (I&A) workshop, and PI Planning.

Business Owners – Are a small group of stakeholders who have the business and technical responsibility for fitness for use, governance, and return on investment (ROI) for a Solution developed by an ART. They are primary stakeholders in the ART and actively participate in ART events

Scaled Agile For Lean Enterprises



Should a Project Manager learn Agile?

The impact of Agile on the role of many project managers is likely to be significant. On small, single-team Agile projects, you may not find someone called a “Project Manager”. However:

- ✓ There certainly is a need for someone to coach and mentor the team on Agile Project Management practices.
- ✓ There is also a need for project managers on larger and more complex enterprise-level projects but even at that level, some understanding of Agile is essential.

This “raises the bar” for project managers significantly. It requires project managers to develop a fresh new perspective to see Agile and traditional, plan-driven project management approaches as complementary to each other rather than competitive and to learn how to blend the two approaches in whatever proportions are needed to fit any situation

The PM role in a Lean and Agile world

In the lean and agile world, the project manager does not have an official role

The Scrum practice prescribes distributing the PM role among the Scrum team members. However, there are varying opinions and experiences assigning any combination of the 3 roles to a PM.

1. **The Scrum Master** (The **Scrum Master** does anything possible to help the team perform at their highest level. This involves removing any impediments to progress, facilitating meetings, and doing things like working with the product owner to make sure the product backlog is in good shape and ready for the next sprint)
2. **The product Owner** (The **Product Owner** is typically a project's key stakeholder. Part of the product owner responsibilities is to have a vision of what he or she wishes to build, and convey that vision to the scrum team. This is key to successfully starting any agile software development project. The agile product owner does this in part through the product backlog, which is a prioritized features list for the product)
3. **The development team member** (Responsible for the project's creation and delivery. The team members will normally be comprised of developers, QA, and documentation. They are responsible for planning, design, development, testing, and project delivery)

The Scaled Agile Framework (SAFe) practice lists the PM as a potential for the Release Train Engineer (RTE). Other agile practitioners describe the PM as a coach and facilitator

Project Manager as a ScrumMaster

- Authority: The traditional project manager role moves from a command-and-control, hierarchical position to a servant-leader or facilitator position
- Requirements: The product owner assumes the responsibility for ensuring the requirements are defined
- Work Assignments: The team takes ownership and accountability for meeting the project and team goals
- Managing Stakeholder Expectations: Product owner provides direction and leadership to the team
- Leadership and Support: The Scrum Master serves the product owner and the team so that they are better able to do their jobs by assisting them, facilitating creativity and fostering empowerment
- Removes Impediments: The Scrum Master helps remove obstacles and support the team

Agile Certified Professional

Become a PMI Agile Certified Professional (PMI-ACP):

General Project Experience	Project Experience	Training	Examination
2,000 hours working on project teams within the last five years. The PMP certification satisfies this requirement	1,500 hours working on agile projects within the last three years	21 hours of training	Pass a 200-question test

The PMI Agile Certified Practitioner (PMI-ACP)® formally recognizes your knowledge of agile principles and your skill with agile techniques. It will make you shine even brighter to your employers, stakeholders and peers

The PMI-ACP spans many approaches to agile such as Scrum, Kanban, Lean, extreme programming (XP) and test-driven development (TDD.) It will increase your versatility, wherever your projects may take you.

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INT422 IT Project Management

Unit III

Agile Processes: Scrum

Lecture 17

Agile Processes: Scrum

- **Scrum**
- **Sprint**
- **Product backlog**
- **Sprint backlog**
- **Sprint review**
- **Retro perspective**

Introduction

- The two dominant Agile approaches are Scrum and eXtreme Programming (XP).
- XP was arguably the first method deemed to be “Agile”.
- We will start with Scrum – very popular and in very wide use today!

Project Management Emphasis based on a Standard 30-day Sprint

- **Scrum:** a definite project management emphasis.
- **Scrum Master:** A **Scrum project** is managed by a Scrum Master, who can be considered as much a consultant or coach as a manager.
- **Sprint.** Scrum has a fundamental 30-day development cycle called a **Sprint**, preceded by
 - **pre-Sprint** activities and **post-Sprint** activities.
- **Daily Scrum:** A short (less than 30 minutes) daily Scrum Meeting allows the team to monitor status and communicate problems.

Product Backlog for Planning

- **Project planning** is based on a **Product Backlog**, which contains
 - functions and
 - technology enhancements
- envisioned for the project.
- Two meetings are held –
 - one to decide the **features for the next Sprint** and
 - the other to **plan out the work**.

Scrum and Scalability

- Scrum: one of the **few agile methods** used to **scale up** for larger projects.
- How done?
 - Accomplished the same way as organizations handle integrated product teams.
 - Individual Scrum team coaches - part of a higher echelon team of coaches spanning several products.
 - This provides for communications to avoid conflicting development issues

Scrum - Queues

- Product Backlog → Sprint Backlog → Sprint → Working increment of the Software
- **Scrum** uses **lightweight queue-based management** and work-breakdown mechanisms.
- **Product Backlog queue:** a low-tech customer-managed queue of demand requests for products.
- .
- **Sprint:** At launch time, a Sprint (30-day time-boxed iteration) does **just-in-time planning**
- **Sprint Backlog:** queue for Sprint work-mgmt.

Scrum - Management

- **Daily Scrum:** Very notable and very visible
- Is a **daily standup**,
 - **except** that it is the **team** that is participating and sharing coordination information **not** a **central project manager**.
- **Scrum Master**
 - holds daily scrum and
 - acts more as a **facilitator** and **runs interference** for the core team when **blocks** or **issues** arise. (Kennaley, SDLC 3.0, p. 31)

Core Roles

- Three **core roles** and a range of **ancillary roles**
- **Core roles:**
 - **Core roles** are those **committed to the project** in the Scrum process
 - Core roles: those **producing the product**

Core Roles – Product Owner

- The Product Owner represents **stakeholders** and is the **voice of the customer**.
- Product Owner is **accountable** for ensuring that the team **delivers value** to the business.
- **Product Owner**
 - **writes** customer-centric items (typically **user stories**),
 - **prioritizes** them, and
 - **adds** them to the **product backlog**.

Note:

- Scrum teams should have **one Product Owner**.
- May also be a member of the development team
- Not recommend this person be Scrum Master.

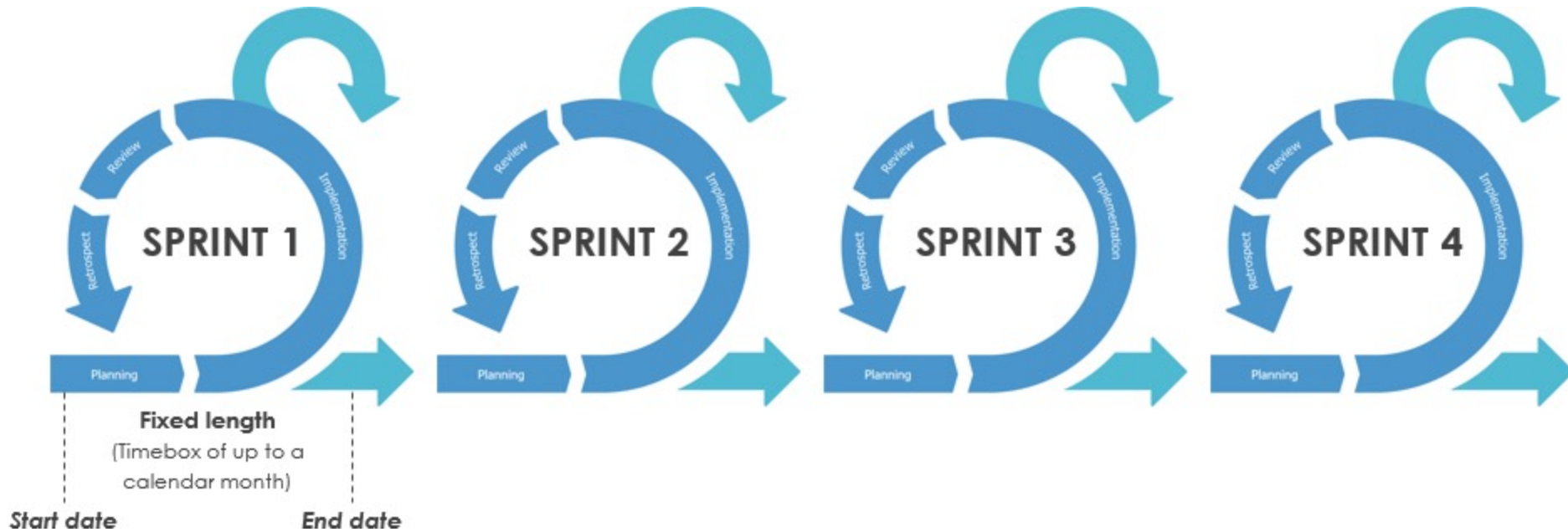
Core Roles – Development Team

- The Development Team is responsible for **delivering potentially shippable product increments** at end of each Sprint.
- Team = 3–9 people with cross-functional skills.
- Team does actual work
 - (analyze, design, develop, test, technical communication, document, etc.).
- Team is **self-organizing**, even though they may interface with project management organizations (PMOs).

Core Roles – Scrum Master

- **Scrum** is facilitated by a Scrum Master –
- Accountable for **removing impediments** for team to deliver sprint goal / deliverables.
- **Scrum Master is not the team leader**, but acts as a **buffer** between the team and any distracting influences.
- Scrum Master ensures **process** is used as intended.
- Scrum Master is the **enforcer of rules**.
- Scrum Master's role: **protect** the Team and keep it **focused** on the tasks at hand.

Sprints



The Sprint (1 of 4)

- **Sprint:** basic unit of development in Scrum.
- **Sprint duration:** one week to one month;
- **“Time Boxed”** effort of a constant length.

- Each sprint:
- Preceded by a **planning meeting**,
 - where the **tasks** for sprint are **identified** and an
 - **estimated commitment for the sprint goal** made, and followed by
 - a **review or retrospective meeting**, where the progress is reviewed and lessons for the next sprint are identified.

The Sprint (2 of 4)

- During Sprint, **team creates finished portions** of a product. (an **increment**)
- **Features** going into a Sprint come from the ***product backlog***: a prioritized list of reqments.
 - **Which** **backlog items** go into sprint (**sprint goals**) determined during Sprint Planning Mtg.
- **Sprint Goal**
 - sets up **minimum success criterion** for the Sprint and
 - **keeps the team focused** on the broader picture rather than narrowly on the task at hand.

Inputs from Executives,
Team, Stakeholders,
Customers, Users

SCRUM


Scrum
Master


Daily Scrum
Meeting

Every
24 Hours

1-4 Week
Sprint


Sprint Review


Finished Work


Sprint
Retrospective



Product Owner



The Team



Product
Backlog

Team selects
starting at top
as much as it
can commit
to deliver by
end of Sprint

Sprint
Planning
Meeting

Task
Breakout

Sprint
Backlog

Sprint end date and
team deliverable
do not change



The Sprint (3 of 4)

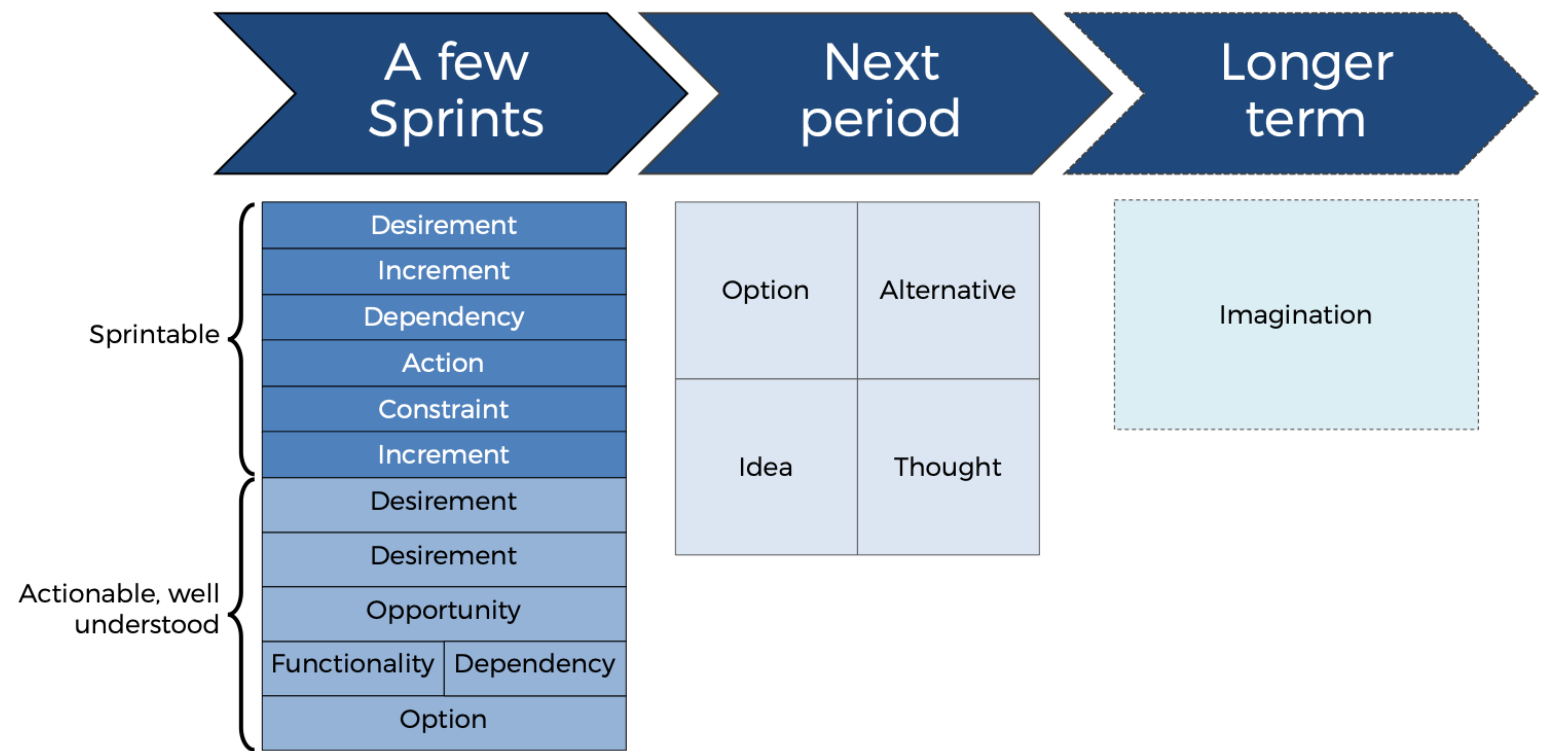
- The **team** then determines **how many selected items** can be **completed** during the next sprint.
- These then go into the **Sprint Backlog**.
- **Sprint Backlog** is **property** of the development team, During a sprint, **no one is allowed to edit the sprint backlog except for development team**.
- **Development: time-boxed**; Sprint **must** end on time;
- Requirements not completed for any reason?
are omitted and **returned** to **Product Backlog**.
- When Sprint is done, team **demonstrates** software.

The Sprint (4 of 4)

- Scrum enables **self-organizing** teams
- Encourages **co-location** of all team members,

Artifact: Product Backlog

Product Backlog as a roadmap



(Gunther Verheyen – Ullizee-Inc)

Artifact: Product Backlog

- **Product backlog** is an ordered list of "requirements" that is maintained for a product
- Contains Product Backlog Items **ordered** by the Product Owner based on
 - considerations like risk,
 - business value,
 - dependencies,
 - date needed, etc.
- **Features** added to backlog commonly written in story format
- The product backlog is the “**What**” that will be built, sorted in the relative order it should be built in.
 - Is open and editable by **anyone**,
 - **Product Owner is ultimately responsible** for **ordering the stories** on the backlog for the Development Team.

Artifact: Product Backlog

- The product backlog contains rough estimates of both business value and development effort, these values are often stated in **story points** using a rounded **Fibonacci** sequence.
- Those **estimates** help the Product Owner to gauge the timeline and may influence ordering of backlog items.
 - Example, if the “add spellcheck” and “add table support” features have the same business value, the one with the smallest development effort will probably have higher priority, because the Return on Investment is higher.

Artifacts – The Product Backlog 2

- **Product Owner:** responsible for **the product backlog** and the **business value** of each item listed.
- **Development Team:** responsible for the **estimated effort** to complete each backlog item.
- Team contributes by estimating Items and User-Stories, either in “**Story-points**” or in “**estimated hours.**”

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- **Scrum**
- **Sprint backlog**
- **Sprint review**
- **Retro perspective**

Sprint Backlog

Sprint Backlog			
Forecast	To-Do	In-Progress	Done
Fix My Profile 5		aliquip	ipsum duis sit ipsum
Filter Service Tickets 8	dolor ipsum culpa	vale culpa	aliquip
Quick Tips 3	ipsum sit duis duis		

Artifacts: Sprint Backlog

- **Sprint Backlog:** list of work the Development Team must address during the next sprint.
- List derived by selecting stories/features from the top of the product backlog until the Development Team feels it has enough work to fill the sprint.
- **Thinking:** This is done by the Development Team asking "Can we also do this?" and adding stories/features to the sprint backlog.
- **History:** Development Team should note **velocity** of previous Sprints (total story points completed from each of the last sprints stories) when selecting stories/features for the **new sprint**.
- Use number as **guide** for "effort" they can complete.

Artifacts: Sprint Backlog

- **Stories/features:** broken down into **tasks** by Development Team
- Should normally be between **four and sixteen hours** of work.
- With this level of detail the Development Team understands exactly what to do, and potentially, anyone can pick a task from the list.
- **Tasks** on sprint backlog are **never assigned**; tasks are **signed up for** by team members during **daily scrum**, according to **priority** and **member skills**.
- Promotes **self-organization** of Team, and **developer buy-in**.
- **Sprint backlog is property of Team**, and all included **estimates** are provided by the Development Team.

Artifacts - Increment

- The **"increment"** is **sum** of all Product Backlog Items completed during a sprint **and** all previous sprints.
- At end of a sprint, Increment must be done according to **Scrum Team's definition** of done.
- The increment must be in **usable condition** regardless of whether the Product Owner decides to actually release it.

Artifacts: Burn Down

- The sprint **burn down chart** is a publicly displayed chart showing **remaining work** in the sprint backlog.
- Updated every day; gives a simple view of the sprint progress.
- Other types of burn down:
- **Release burn down chart:** shows amount of work left to **complete** the target commitment for a Product Release
 - This normally spans multiple iterations
- **Alternative Release burn down chart:** basically does the same, but clearly shows scope changes to Release Content, by resetting the baseline.
 - This should not be confused with an earned value chart.

Scrum Terminology

Scrum Team: Have already discussed

- **Product Owner:**
- **Scrum Master:**
- **Development Team:**
- **Product Backlog**
- **Sprint Backlog**
- **Sprint**

More Terminology Used in Scrum:

- **Sprint burn down chart:** Daily progress for a Sprint over the sprint's length.
- **(User) Story:** A feature added to the backlog is commonly referred to as a story; has a specific suggested structure.
- Done so development team can identify user, action and required result in a request; simple way of writing requests anyone can understand.
- **Example:** As a wiki user I want a tools menu on the edit screen so that I can easily apply font formatting.

More Terminology Used in Scrum:

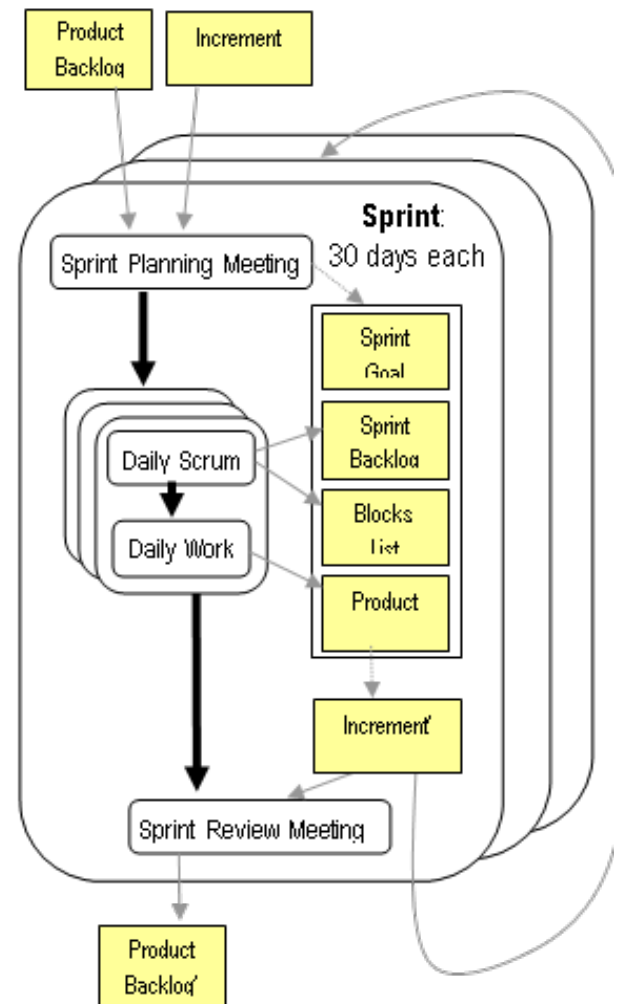
- A **story** is an
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More Terminology Used in Scrum:

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- This is a **guideline** for the team and assists them in understanding **how many stories** they can do in a sprint.
- **Impediment:** Anything that prevents a team member from performing work as efficiently as possible.



Sprint review

- The Sprint Review is the second to last event of the Sprint and is time boxed to a maximum of four hours for a one-month Sprint. For shorter Sprints, the event is usually shorter.

The Sprint Review may include the following elements and more:

- Attendees include the Scrum Team and key stakeholders invited by the Product Owner;
- Members of the Scrum Team explain what Product Backlog items have been “Done” and what has not been “Done”;
- The [Developers](#) discuss what went well during the Sprint, what problems it ran into, and how those problems were solved;
- The Developers demonstrate the work that it has “Done” and answers questions about the Increment;

Sprint review

The Sprint Review may include the following elements and more:

- The Product Owner discusses the Product Backlog as it stands. He or she projects likely target and delivery dates based on progress to date (if needed);
- The entire group collaborates on what to do next, so that the Sprint Review provides valuable input to subsequent [Sprint Planning](#);
- Review of how the marketplace or potential use of the product might have changed what is the most valuable thing to do next; and,
- Review of the timeline, budget, potential capabilities, and marketplace for the next anticipated releases of functionality and capability of the product.

Sprint review

- **Velocity:** The total effort a team is capable of in a sprint. The number is derived by adding all the **story points** from the last sprint's stories/features.
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INT422 IT Project Management

Unit III

Roles in Scrum

Lecture 19

Agile Processes: Scrum

- **Scrum - Roles**



Scrum Roles

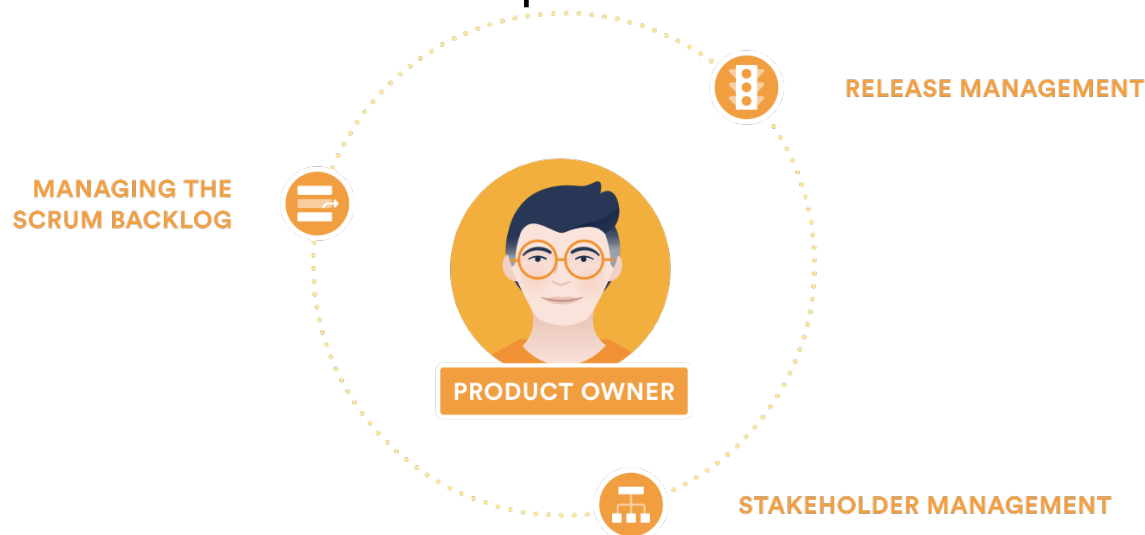
- **A Scrum team is made up of three roles: the Scrum master, the product owner, and development team members.**
- While this is pretty clear, what to do with existing job titles can get confusing. Many teams ask if they need to change their titles when adopting scrum.

Scrum roles vs. job titles

- The three scrum roles describe the key responsibilities for those on the scrum team. They aren't job titles. This means that any job title, even your existing ones, can perform one of the roles.
- Because the essence of scrum is empiricism, self-organization, and continuous improvement, the three roles give a minimum definition of responsibilities and accountability to allow teams to effectively deliver work.
- This allows teams to take responsibility for how they organize and to keep improving themselves.

What are the three scrum roles?

- A Scrum team consists of three distinct roles: the Scrum master, the product owner, and development team members. While there is only one Scrum master and one product owner, there are generally several development team members.
- Scrum teams are small; the Scrum Guide recommends ten or less total members to make sure communication and productivity is optimal. Let's take a closer look at what the responsibilities are for each of these roles.



1. Scrum master

- A Scrum master is the person responsible for making sure a Scrum team is operating as effectively as possible with Scrum values.
- This means they keep the team on track, plan and lead meetings, and work out any obstacles the team might face. Scrum masters might also work in a larger role within an organization
- Scrum can look different from organization to organization and team to team, making the specific tasks of a Scrum master varied. Broadly, however, a Scrum master might have the following responsibilities



1. Scrum master

- Facilitate daily Scrum meetings (also called “daily standups”)
- Lead sprint planning meetings
- Conduct “retrospective” reviews to see what went well and what can be improved for the following sprint
- Keep a pulse on team members, through individual meetings or other means of communication
- Manage obstacles that arise for the team by communicating with stakeholders outside of the team

2. Product owner

- A product owner makes sure the Scrum team is aligned with the goals of the overall product that the team is contributing to.
- They understand the business needs of the product, like customer expectations, and market trends. Because they have to understand how the Scrum team fits into bigger picture goals, product owners usually stay in touch with product managers and other stakeholders outside the team.

2. Product owner

- **Product owners might find themselves with the following responsibilities:**
- **Manage the product backlog by ordering work by priority**
- **Set the product vision for the team**
- **Communicate with external stakeholders and translate their needs to the team**
- **Make sure the team is focused on hitting product needs through communication and evaluating progress**

3. Development team

- **A development team is composed of the professionals who do the hands-on work of completing the tasks in a Scrum sprint.**
- **This means development team members can be computer engineers, designers, writers, data analysts, or any other role needed to reach sprint goals.**
- **The development team doesn't just wait for orders; they usually collaborate to map out goals and plans for achieving them.**

3. Development team

- **Not all development team members will always have the same responsibilities. For example if you're updating a website, you might have a front-end engineer, UX designer, copywriter, and marketing professional all working on the same Scrum team. The responsibilities of a development team will also depend on the end goals of the Scrum team. Broadly speaking however, you might find that a development team can be tasked with the following:**
 - **Help in sprint planning and goal setting**
 - **Lend expertise to program, design, or improve products**
 - **Use data to find best practices for development**
 - **Test products and prototypes, plus other forms of quality assurance**

Knowledge about SCRUM

How much does a Scrum master make?

- Professionals who are dedicated Scrum masters make an average salary of \$97,757 in the US, according to Glassdoor data as of August 2021 [\[4\]](#). Senior Scrum masters make an average base salary of \$115,167, Glassdoor reports.

Should Scrum roles be merged?

- Distinguishing the three roles in Scrum—Scrum master, product owner, and development team—is meant to heighten the transparency, efficiency, and adaptability of a team. In classic Scrum teams, the three roles are considered vital to success.

Is a Scrum master a project manager?

- Scrum masters can be considered a type of [project manager](#), but their definitions differ in some basic ways. While a Scrum master keeps a Scrum team on track and in sync with Scrum principles, a project manager is more concerned with elements of a project's success—things like staying within budget, meeting timelines, and hitting goals.

More Terminology Used in Scrum:

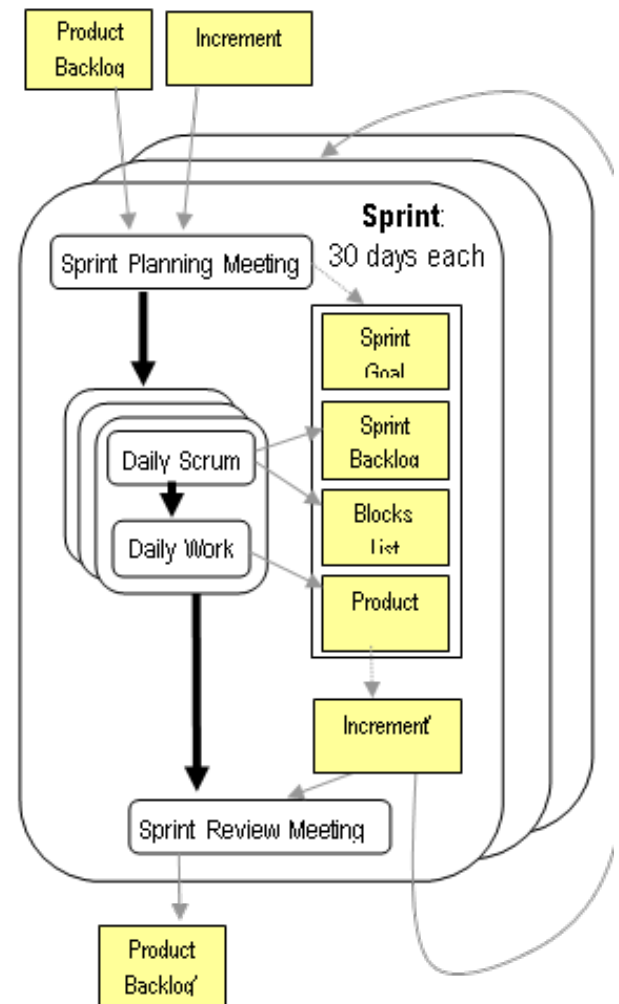
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INT422 IT Project Management

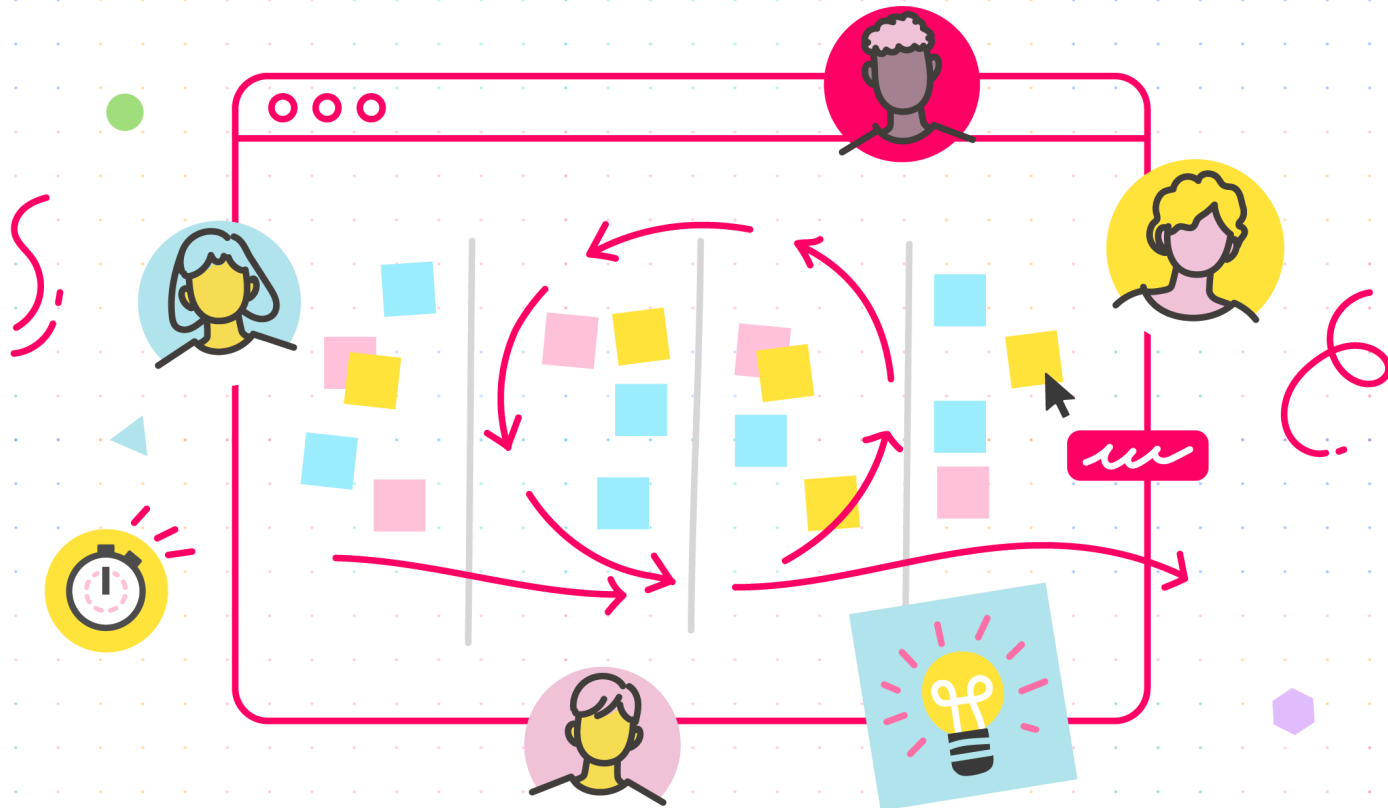
Unit III

Best practices of Scrum

Lecture 20

Agile Processes: Scrum

- **Best Practices Scrum**



Best Practices Scrum

1. Adhere to Agile Scrum Ceremonies

The Scrum ceremonies include:

- Daily scrum (aka daily standup)

- Sprint planning

- Sprint review

- Sprint retrospective

Why Is It Best Practice?

You should adhere as closely as possible to Agile Scrum ceremonies, which are put in place to ensure every step of the process goes smoothly. Omit one, and you will face problems later in the project cycle.

For instance, without the daily standup, your team could flounder without direction, with team members waiting for instruction on how to proceed.

This is especially important in mission-critical projects like software development.

2. Get Scrum Roles Identified Quickly

- Scrum roles are titles that delineate who is responsible for what during the process. The three main roles are:
- *Product Owner* — The Product Owner is the boss.
- *Scrum Master (or Scrum Leader)* — The Scrum Master is the person responsible for making sure the team adheres to Agile Scrum ceremonies and collaborates efficiently.
- *Developer* — Developers are team members who take the Tasks/User Stories and develop the Sprint planning

3. Make Sure to Prioritize User Stories Before Your Project Gets Started

- User Stories (or Tasks) are the most important part of an Agile Scrum Board. These are the actionable items that must get done — including features, enhancements, and fixes for your product or service.
- But it's not as simple as cherry-picking what you want to do and placing the tasks on the board. These tasks go into a kind of well, called the Product Backlog.
- Then they are distributed by a Product Owner in order of priority, team resources required, or other relevant factors such as company mission or directive. Finally, they go into the Sprint Backlog.

3. Make Sure to Prioritize User Stories Before Your Project Gets Started

- The Sprint backlog is the responsibility of the Scrum Master and the development team. This is a subset of the Product Backlog, and these are the Tasks/User Stories the development team decides can get done in a reasonable Sprint.
- Think of the Product Backlog as a big picture and the Sprint Backlog as a smaller picture that is more likely to get done.

Make sure you prioritize all of your work before you start a project.

MURAL TEMPLATE

Prioritize projects based on value for your user

User Story Map

a template brought to you by your friends at MURAL

INTRODUCTION

User story mapping is a useful tool for planning releases based on how your users will interact with each iteration. Our user story map template can help you plan and prioritize your work based on where it will create the most value for your users.



BEFORE YOU START

- 1. PERSONAS:** Start with the personas at the top of the template. You can add personas relevant to the project ahead of the session and review during the call.
- 2. USER JOURNEY:** Next you will flesh out the steps in the user journey. Use this as an opportunity for the team to resolve discrepancies about the journey and get on the same page.
- 3. BACKLOG:** Finally, sort the user stories into releases based on the user's journey in an open discussion. You can add known stories in advance before the session starts. Import from Jira and add to section. Use the connectors to track dependencies.

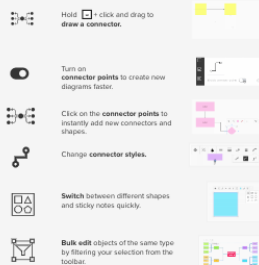
TOOL TIPS

Make plans and put them into action in Jira:

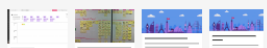
Process outputs from prioritization and planning sessions quickly and efficiently by exporting sticky notes from your mural directly into Jira.

- Import tasks into MURAL from Jira**
To import issues into a mural to plan and prioritize, simply right click on the canvas and choose "Import Jira Issues."
- Export tasks from MURAL to Jira**
After your session in MURAL, right click on a sticky note (or select multiple sticky notes) and choose "Send to Jira."

Create connections at the speed of thought:



RESOURCES



1 PERSONAS

Create descriptions of your users.

- As a team, determine the key users who will interact with the product. Then complete the proto-persona by filling the four areas of the framework.
- Discuss how the system will address their needs and pain points.

IMPORTANT: If you already have personas, remove the frameworks to the right and add them there.

PROTO-PERSONA 1



PROTO - PERSONA 2



RESOURCES



TEAM



LEGEND



2 USER JOURNEY

- Add the broader phases of interaction the user has with the product along the top, e.g. "Log in," "Search," "View content."
- Indicate the specific tasks of interaction, e.g. "Enter password," "Filter list," "Sort content."



Release 1

Date



Release 2

Date



Release 3

Date



3 BACKLOG

- List known stories in advance; or, use this space to park future stories not in any of the releases above.
- Use the connectors to link stories to each other to show dependencies.



4. Focus the Daily Stand Up for a More Efficient Workflow

- The Daily Standup or Daily Scrum is the only time during the day where the team meets to discuss the tasks on the board and calls out problems or obstacles.
- Since you only schedule 15 minutes for these meetings (to keep focus and not waste anyone's time), it's crucial the meetings have an agenda and important issues are prioritized.
- A Scrum Master may want to review the board at the end of the day and come up with actionable agenda items for the next standup.

A good stand-up leaves room for fun, communication, and planning.



Quickly connect members of a team to share priorities and collaborate on roadblocks.

The Ultimate Team Stand-up

Introduction

A stand-up is a short team meeting held either daily or weekly to promote transparency and collaboration. In addition to the project management value for Agile teams, a stand-up is also an opportunity to connect as a team and check-in with everyone.



PEOPLE
4-12



TIME
30 MINS



DIFFICULTY
BEGINNER

Before you start

Customize the categories on the radar "pie slices" in STEP 2 based on your team's focus and alignment. For example, you can label each slice with a department name, like Marketing, Sales, etc. — or each slice could represent a different platform category, like Security, Front-end, Back-end, etc.

Agenda

- 1 Check-in
- 2 What's on your radar?

Resources



1

Check-in

Claim a color by adding your name



Which cat are you today?

Duplicate your colored stick note from above and add it to the cat that best describes how you feel today.

Briefly discuss clusters and outliers.



Snug



Business as usual



Need more coffee



Bottlenecked



No idea what I'm doing



Seen some things



Grawl!



Relaxed



Focused!



Climbing the walls



Deal with it



Pretending to work

brought to you by cats of

2

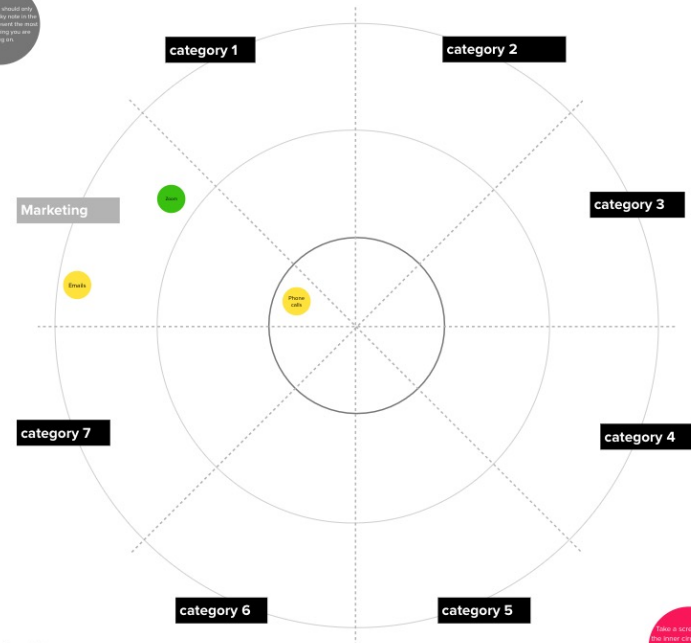
What's on your radar?

Indicate the top activities you're working on for the day or week using the same sticky note color from the first activity.

Prioritize the activities

Sticky notes placed in the center bullseye are higher priority than ones on the outer edges of the circles.

[team name here]



Download for free! (script: Simple) | What's on Your Radar?

LUMA APP AT 10/15/21

LUMA APP AT 10/15/21



5. Run Effective Sprint Retrospective Meetings

- **A Sprint retrospective is a post-mortem collaboration that lets the team go over what was done, what was done right, what went wrong, and what could be done better.**
- **But an ineffective Sprint Retrospective can be just as time-wasting as an ineffective daily standup.**

Click on the image above to learn more about our Product School-certified Retrospective template

MUR

AL

Template Certified by Product School

PRODUCT SCHOOL

Evaluate and evolve your work

Retrospective Template

INTRODUCTION

This retro template provides space for you and your team to collaboratively reflect on a recent project, sprint, or time period and identify actions for improvement going forward.

Use the template to check in with team members, hear multiple perspectives, celebrate wins, and identify ways to improve collaboration in the future.

This exercise is valuable for teams whether or not they follow other Agile practices.

PEOPLE
2 - 10

TIME
40 minutes

DIFFICULTY
Beginner

RESOURCES



Product Template Retrospectives

The following retrospective meeting Template has been validated by The Group Product Manager at Google and Senior Product Manager at Jira.

Product School



Retrospective

Evaluate and evolve your work

Mural



Make plans in MURAL and put them into action in Jira

Proven advice for successful sprint and planning sessions quickly and efficiently by reporting sticky notes from your mural directly into Jira.

Jira

RESOURCES

Daily Scrum

Backlog Refinement

Sprint Planning

Sprint Review

Sprint Retrospective

Date, Month, Year

Check in



What went well?



What didn't go well?



Actions to take



Check out



0%

+100%

Use an Agile Scrum Board to Make Organization a Priority

Incorporating these best practices into your Agile Scrum Board will lead to a more focused process, help advance your project regardless of size or scalability, and foster more cooperation among your teams.

If the devil is in the details, you don't want to lose control of the details. Following Agile Scrum board processes faithfully will lead to increased productivity and decrease wasted time.

INT422 – Unit 3

Agile Project Management

Dr. S.Markkandeyan

~~Senior~~ Assistant Professor

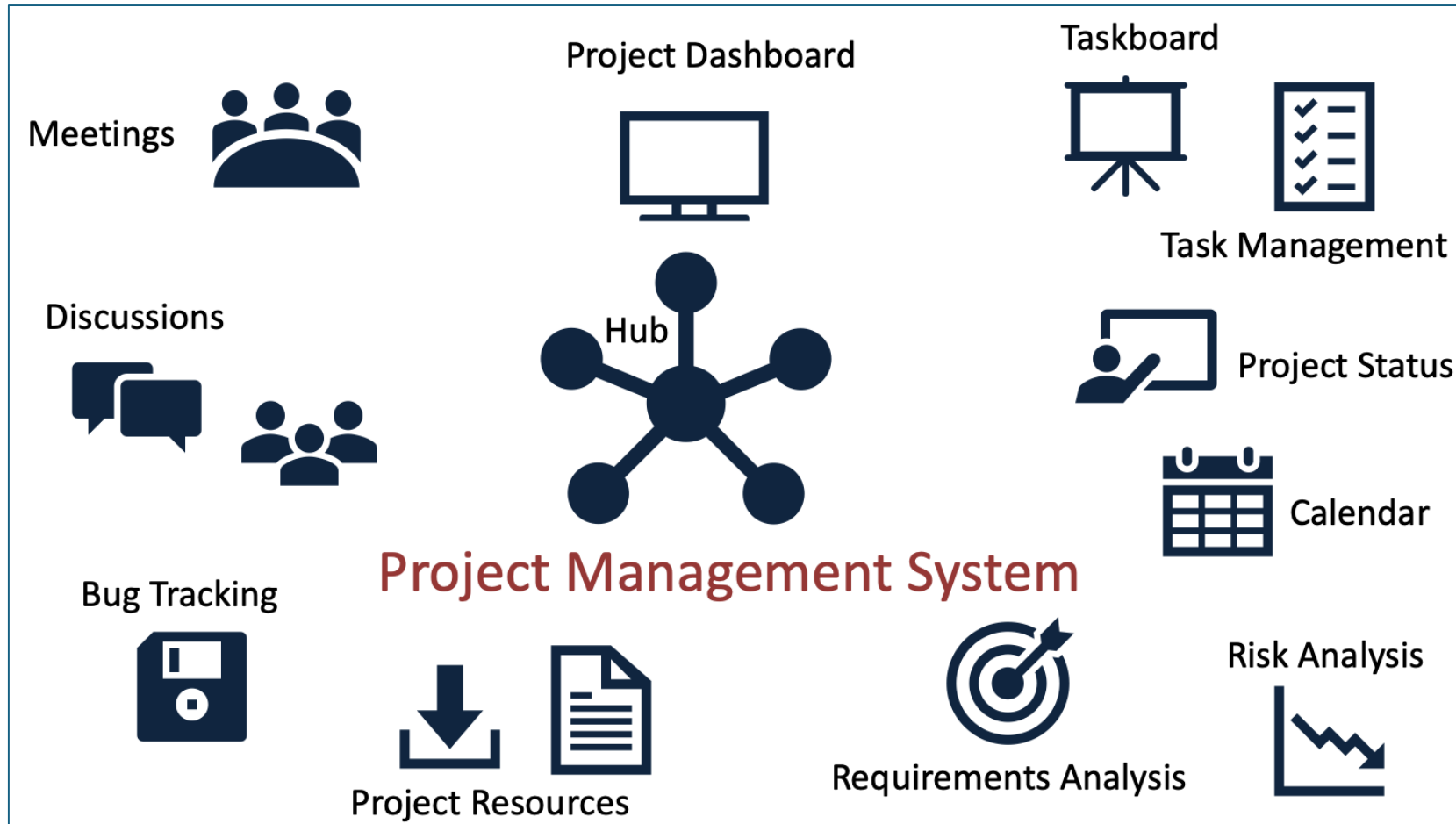
School of Computing SASTRA

Deemed to be University

Thajavur

Art of Project Management

“the application of **knowledge, skills, tools and techniques** to project activities to meet the project requirements”.

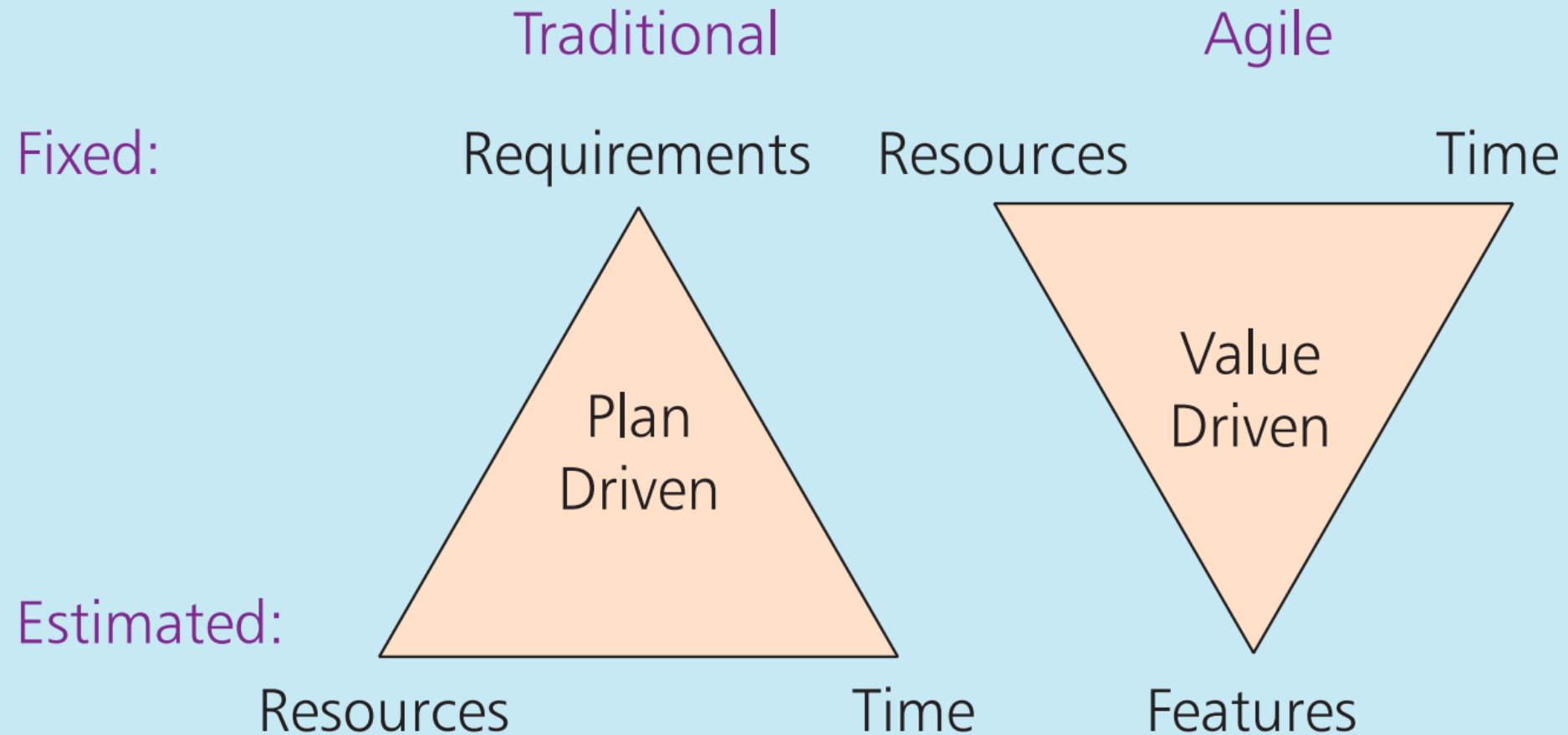


Project Management Phases

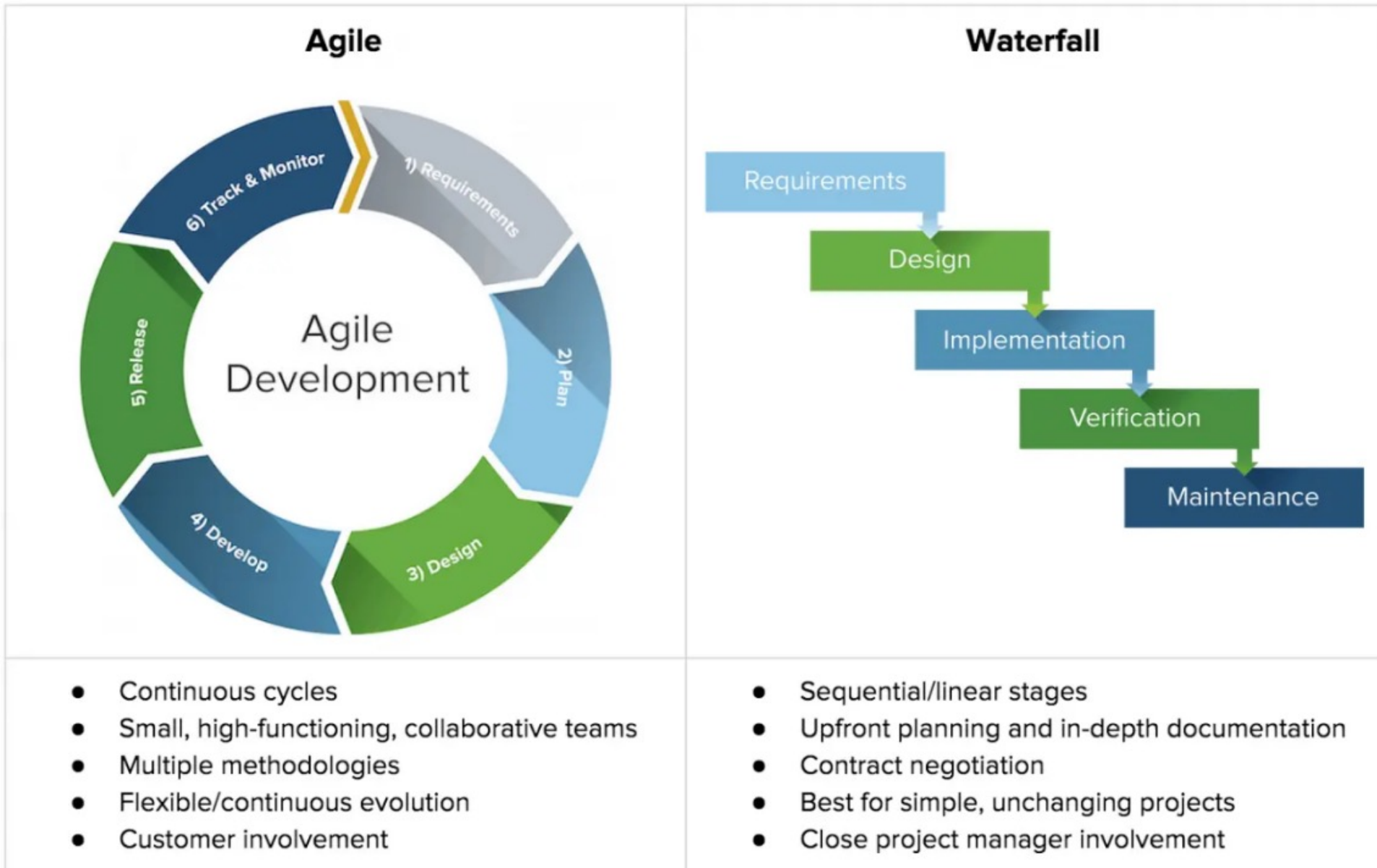
1. Initiation
2. Planning
3. Execution
4. Performance/Monitoring
5. Project close

Agile vs Traditional

The Paradigm Shift



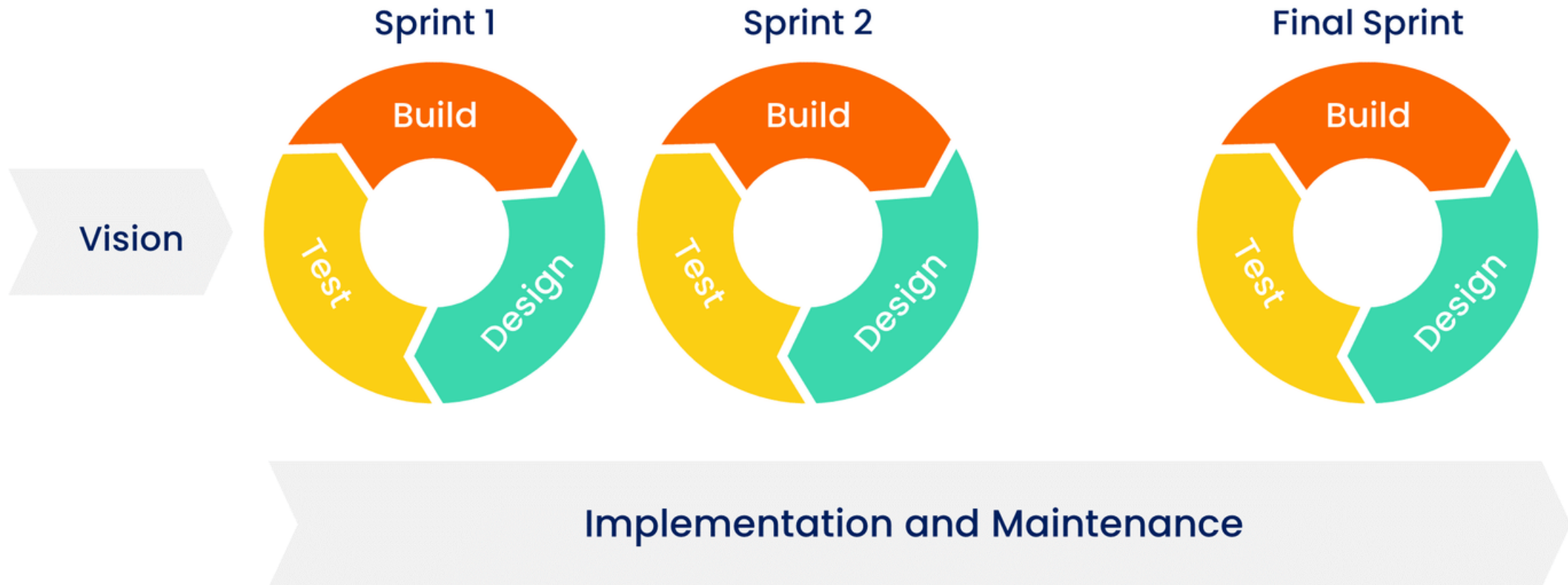
Agile vs Traditional



Agile Phases

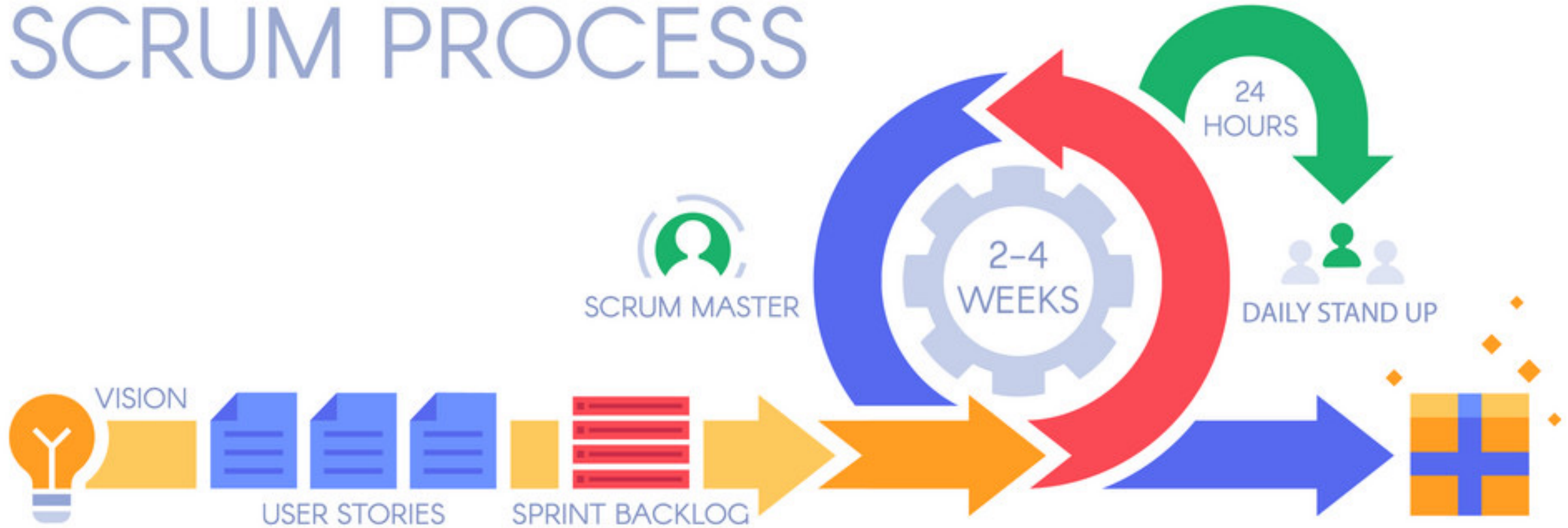


Agile Phases



Agile Phases

SCRUM PROCESS

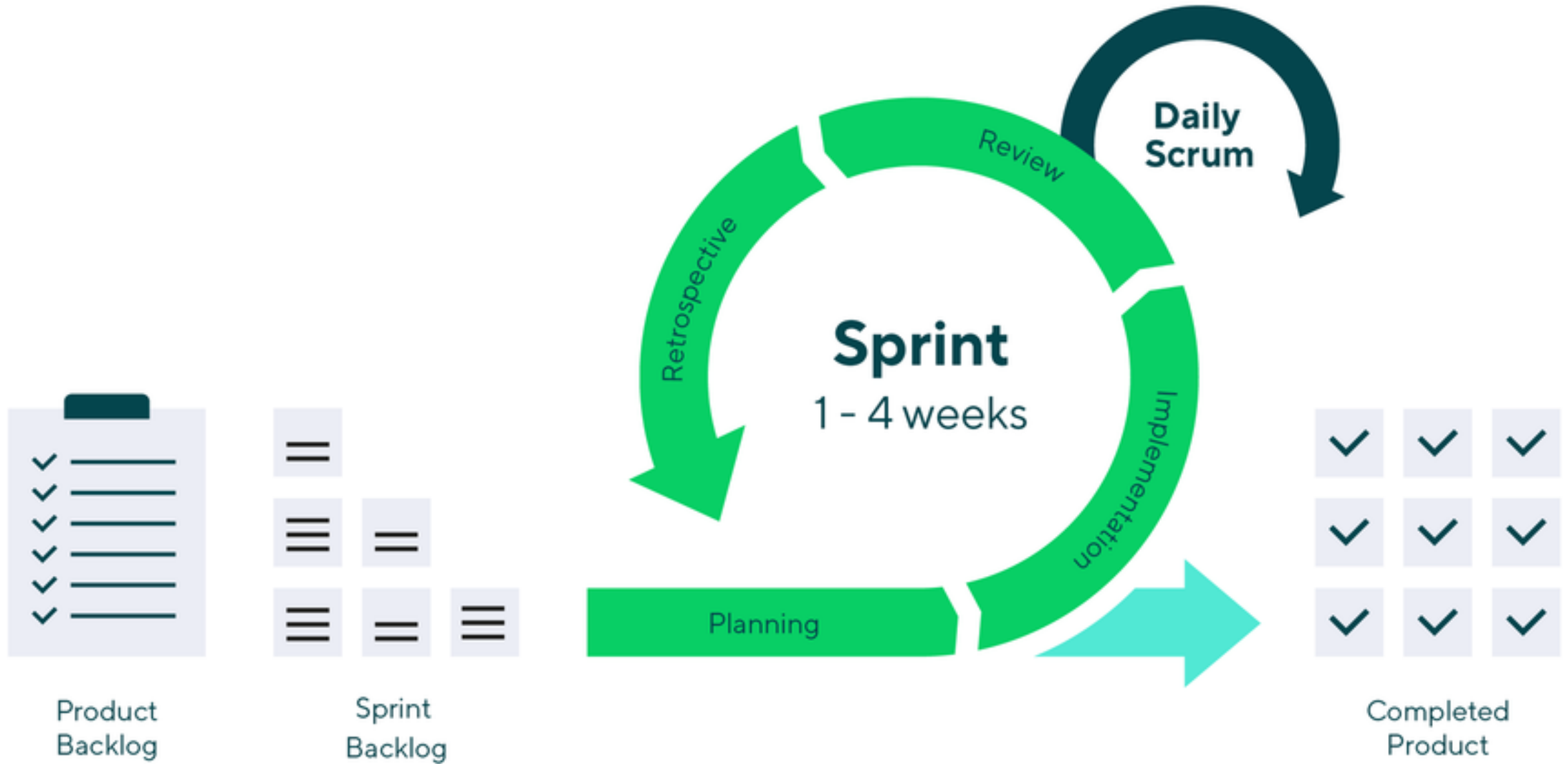


Agile Phases



Example

Agile Phases



Agile Methodology

- Agile Development is a term for **several iterative and incremental** software development methodologies.
- Although each of the agile methods is unique in its specific approach, they all share a common vision and core values.
- There are 2 approaches in agile development, namely:
 1. Lightweight approach
 2. Fuller approach

Agile Methodology

Lightweight approach

- A. Scrum,
- B. Lean software development,
- C. Kanban (process + method),
- D. Continuous Integration,
- E. Continuous delivery,
- F. Extreme programming (XP), and
- G. Feature-driven development (FDD).

Fuller approach (The full approach is used to deploy for more than one team.)

- A. Scrum of Scrums,
- B. Scrum at Scale,
- C. Large scale-scrum,
- D. Scaled agile framework,
- E. Disciplined agile delivery,
- F. Agile unified process

Values of Agile

1. **Individuals and interactions** over processes and tools
2. **Working software** over comprehensive documentation
3. **Customer collaboration** over contract negotiation
4. **Responding to change** over following a plan

12 Principles of Agile

1

Satisfy the customer

Our highest priority is to satisfy the customer through early and continuous delivery of valuable software.

2

Welcome changing requirements

Even late in development. Agile processes harness change for the customer's competitive advantage.

3

Deliver working software frequently

from a couple of weeks to a couple of months, with preference to the shorter timescale.

4

Work together

Business people and developers must work together daily throughout the project.

5

Build projects around motivated individuals

Give them the environment and support they need, and trust them to get the job done.

6

Face-to-face conversation

The most efficient and effective method of conveying information to and within a development team is face-to-face conversation.

12 Principles of Agile

7

Focus on working software

Working software is the primary measure of progress.

8

Promote sustainable development

Agile processes promote sustainable development. The sponsors, developers, and users should be able to maintain a constant pace indefinitely.

9

Ensure technical excellence

Continuous attention to technical excellence and good design enhances agility.

10

Simplicity

The art of maximizing the amount of work not done – is essential.

11

Self-organizing teams

The best architectures, requirements, and designs emerge from self-organizing teams.

12

Reflect and adjustment

At regular intervals, the team reflects on how to become more effective, then tunes and adjusts its behavior accordingly.

Agile Overview

- Agile project management is based on an **incremental, iterative** approach.



Agile Development Life Cycle

Planning

Once an idea is deemed viable, the project team comes together to identify features, prioritize each feature, and assign them to an iteration.

Requirements analysis

Key stakeholders and users meet to identify business requirements that are quantifiable, relevant, and detailed.

Design

The design is prepared from the requirements identified and the team considers what the product or solution will look like, deciding on a test strategy or plan to proceed.

Implementation, coding, development

Coding or developing features, scheduling iterations for deployment.

Testing

Test the code against the requirements to make sure the product is actually solving customer needs. This phase includes unit testing, integration testing, system testing, and acceptance testing.

Deployment

Deliver the product to customers. Once customers start using the product, they may run into new problems that the project team will need to address in future iterations.

Popular Types of Agile Methodologies

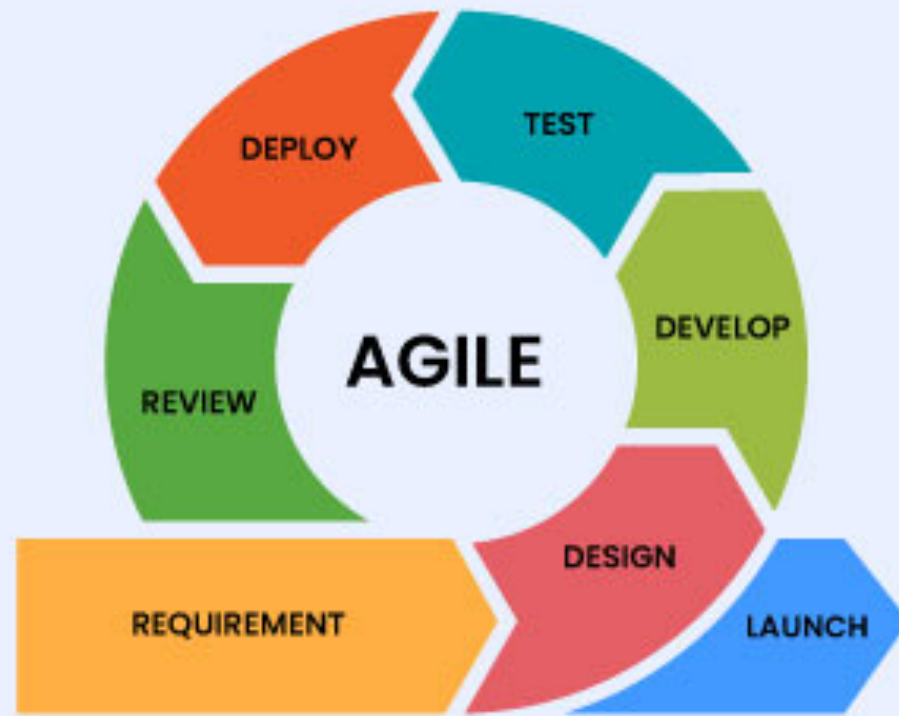
1. Scrum
2. Kanban
3. Lean
4. Extreme Programming
5. Crystal
6. Feature Driven Development (FDD)
7. Dynamic System Development (DSDM)

Benefits of Agile Project Management

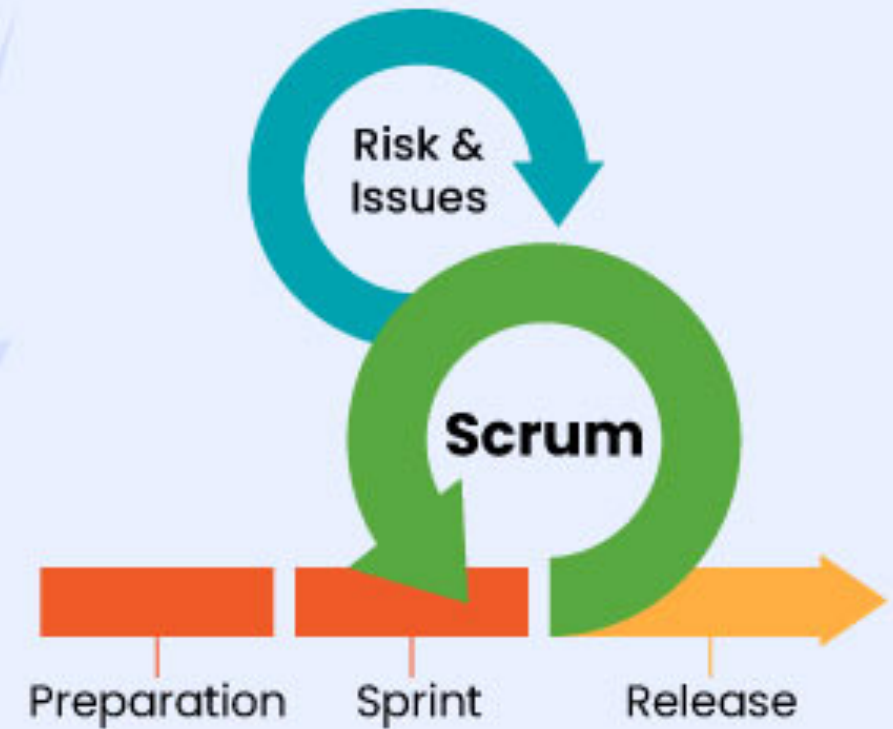
1. Better control of projects
2. Reduced risk
3. Increased Flexibility
4. Higher Customer Satisfaction
5. Enhanced Team Collaboration
6. Efficient Use of Resources

Agile Scrum vs Lean vs DevOps vs ITIL

Agile vs Scrum: Key Differences



VS



Agile Scrum vs Lean vs DevOps vs ITIL

Lean vs Agile: Key Differences



Eliminating waste, Optimizing processes and Maximizing value

VS



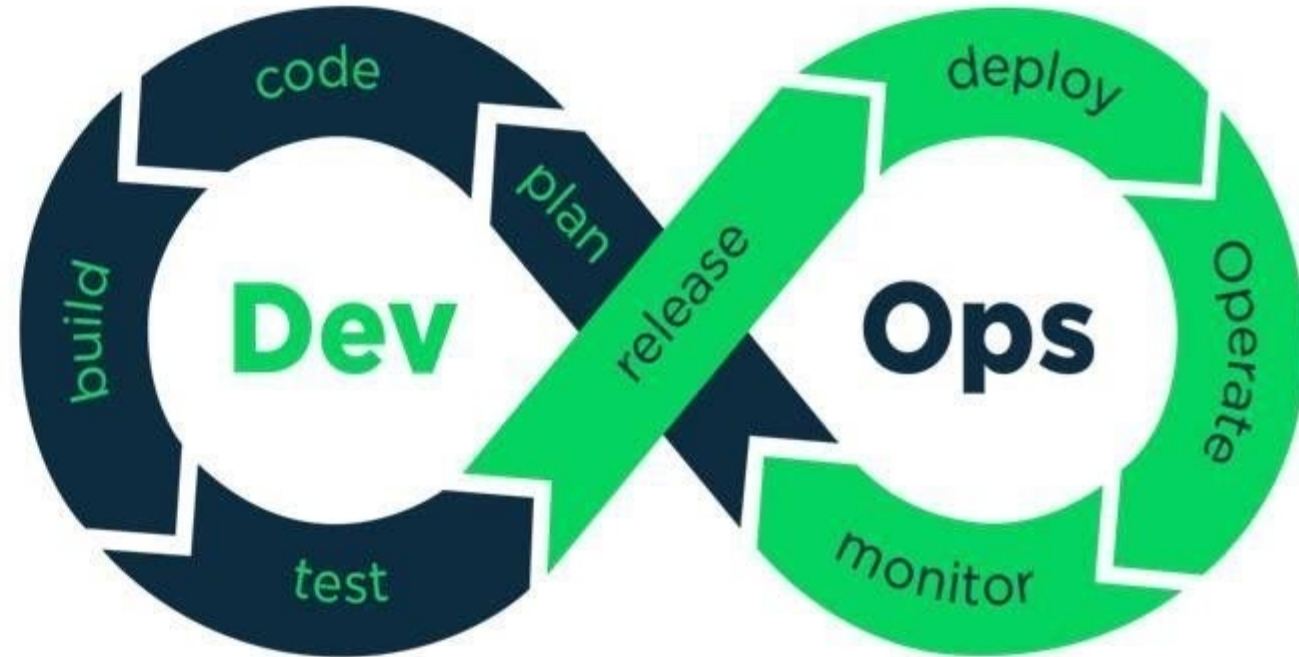
Flexibility and rapid adaption to changes

Agile Scrum vs Lean vs DevOps vs ITIL

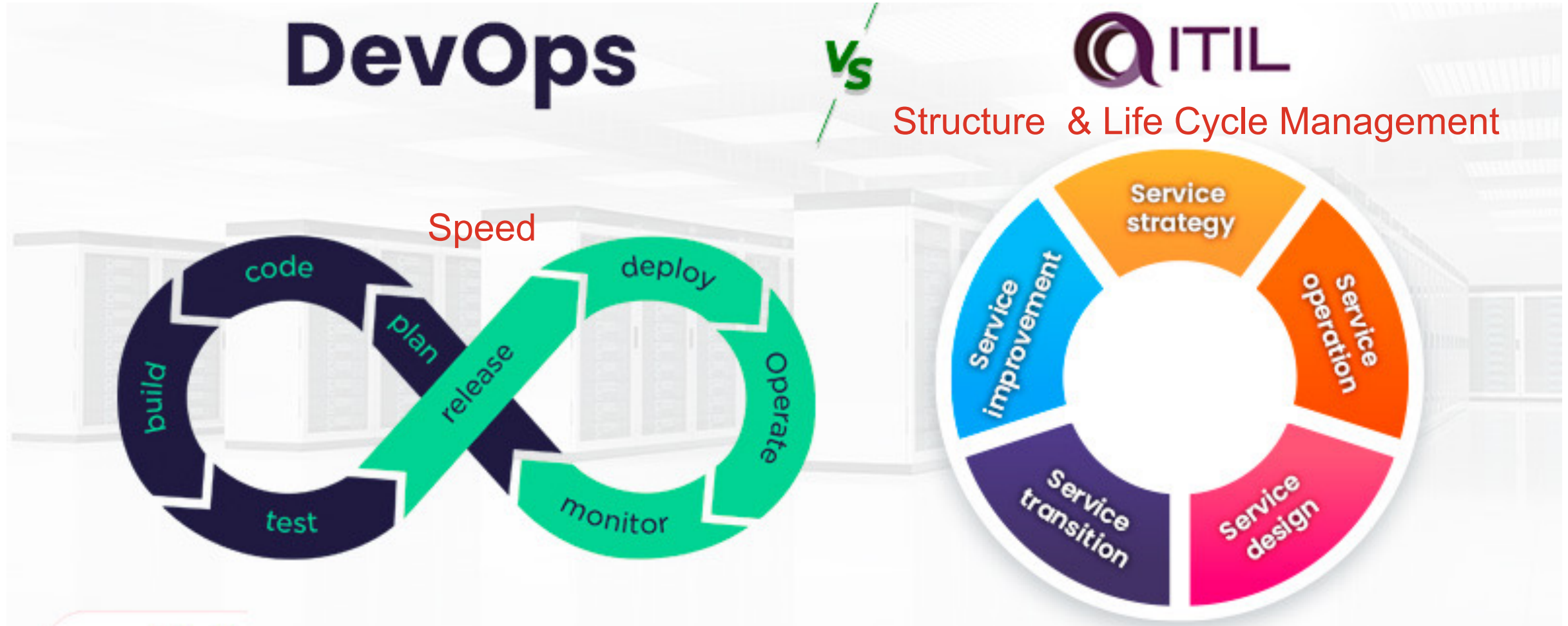
Development



Development and Operations



Agile Scrum vs Lean vs DevOps vs ITIL



1. Scrum

- Scrum is an **iterative and incremental** Agile software development framework providing **continuous development** of the project **in each iteration and adapting changes** for managing software projects and product or application development.
- Scrum enables the **creation of self-organizing teams** and **verbal interaction among all team members**. Not a Defined Process but an Empirical Process.
- In simple terms, Scrum can be defined as an Agile framework for **team collaboration** to build complex projects.

Roles in Scrum Team

1. Scrum Master
2. Product Owner
3. Development Team

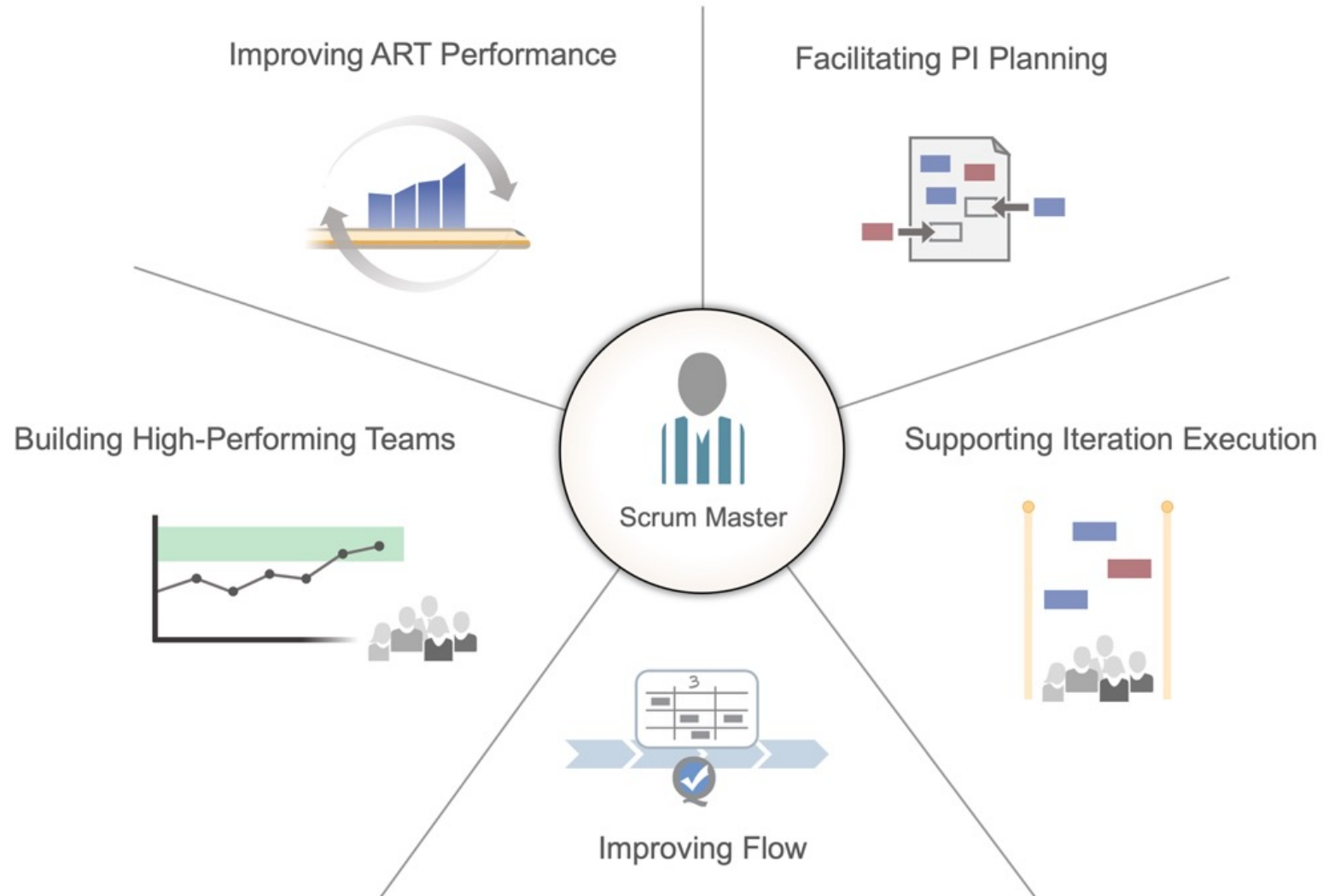
How Agile makes easy?

1. Product Grooming Session
2. Sprint Planning Session
 - What to do, and how to do
 - Daily standup meeting
3. Sprint Review Meeting
4. Sprint Retrospective Meeting

Scrum Terminologies: Product Owner



Scrum Terminologies: Scrum Master

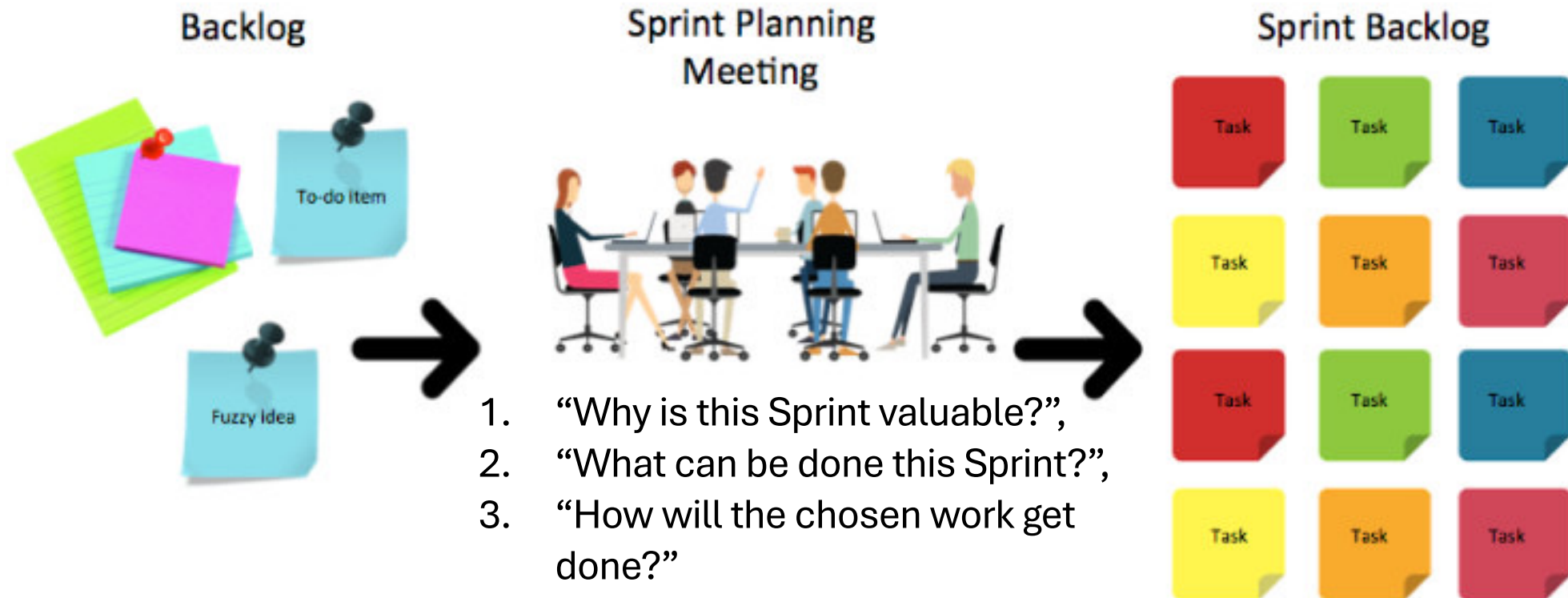


Scrum Terminologies: Sprint



Scrum Terminologies: Sprint Planning

Sprint Planning Meeting



Scrum Terminologies: Daily Scrum

Daily Scrum Meeting



Time box



Same place



Same time



Facilitated by
Scrum Master



Full team
presence



Focus on 3
questions

3 Main Questions:

1. What did I do yesterday?
2. What will I do today?
3. What's in my way?

Scrum Terminologies: Sprint Review

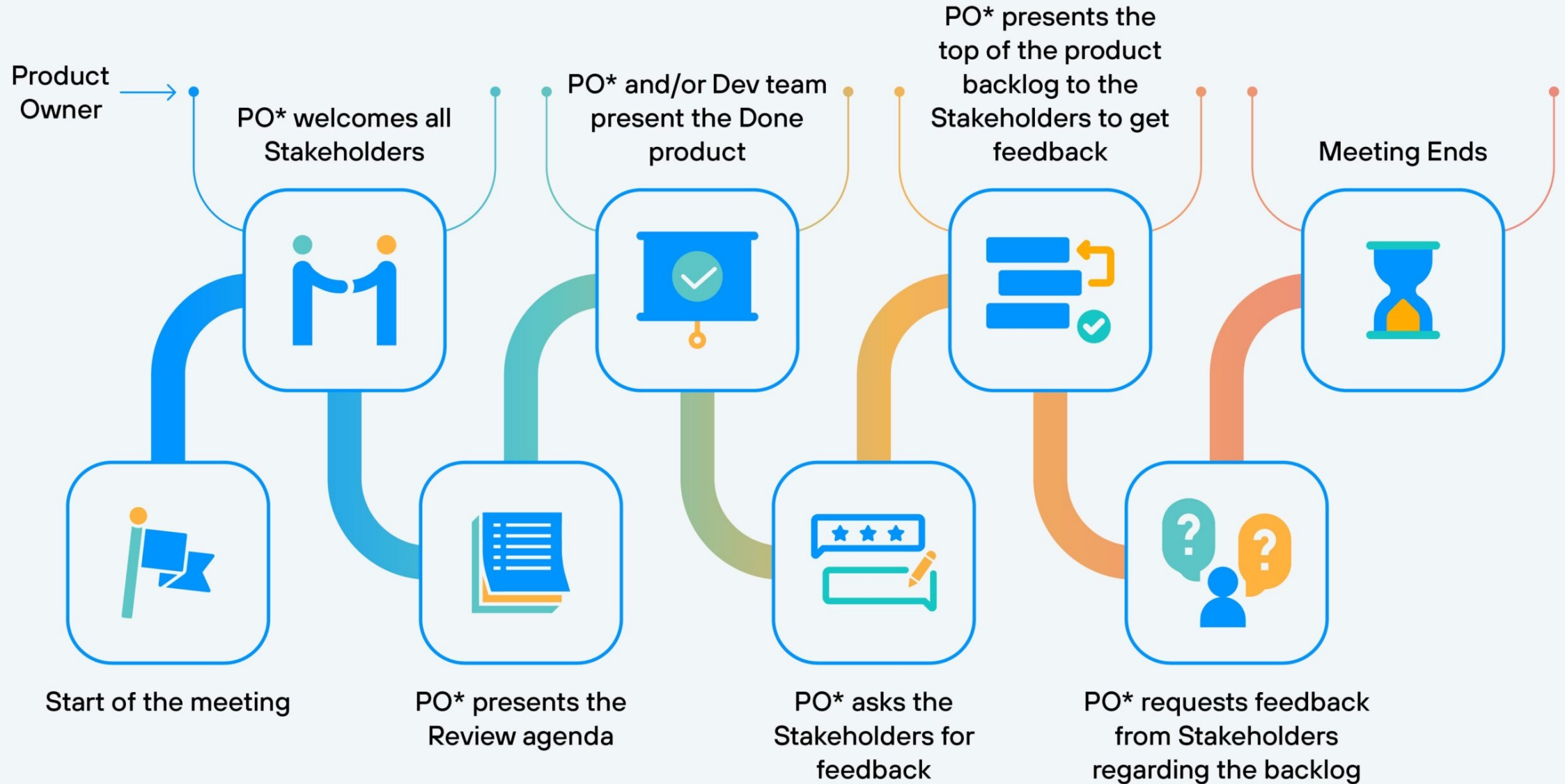
Sprint Review

Meeting at the end of the sprint to check the increment

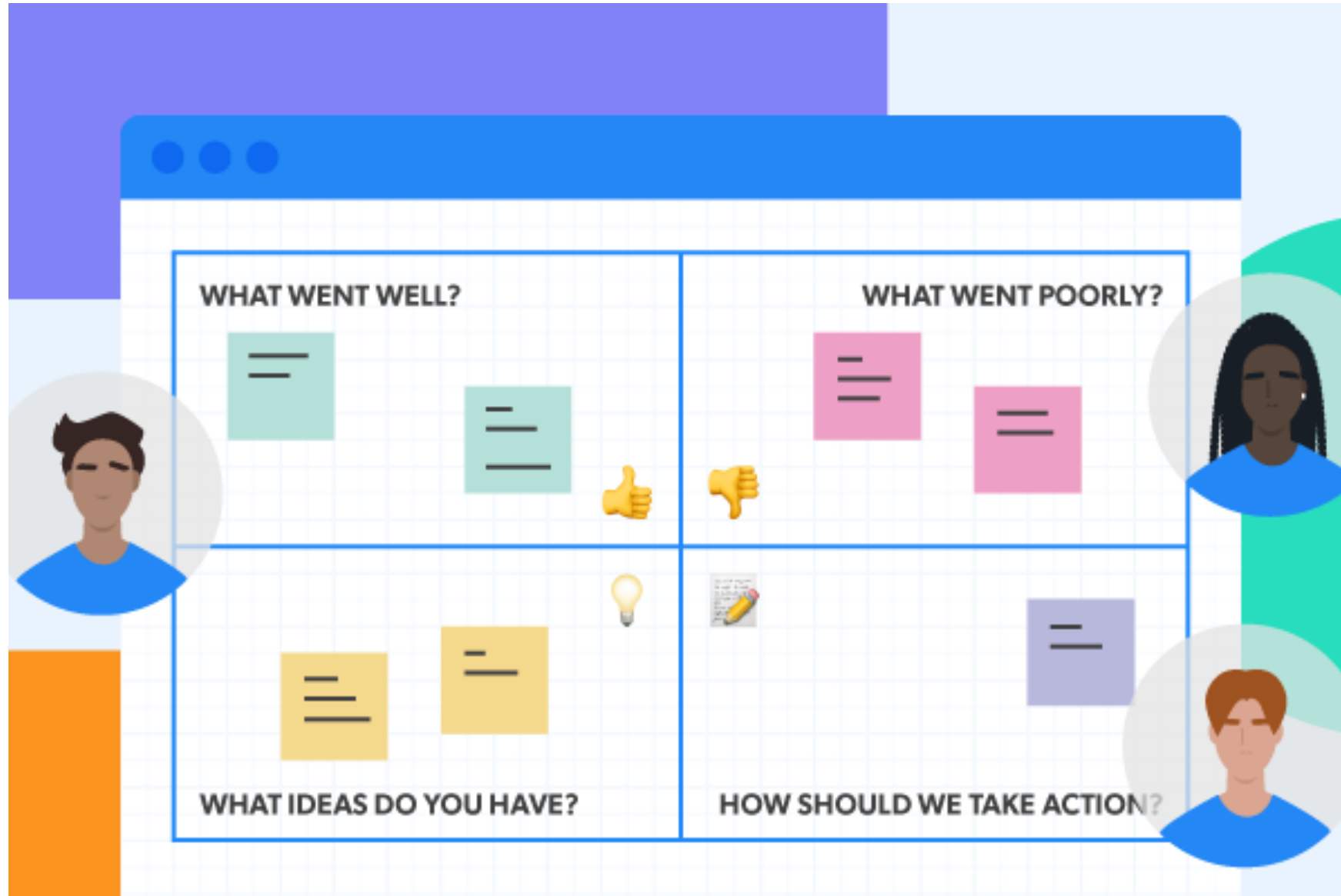


Product owner + Scrum Master + Team + Others

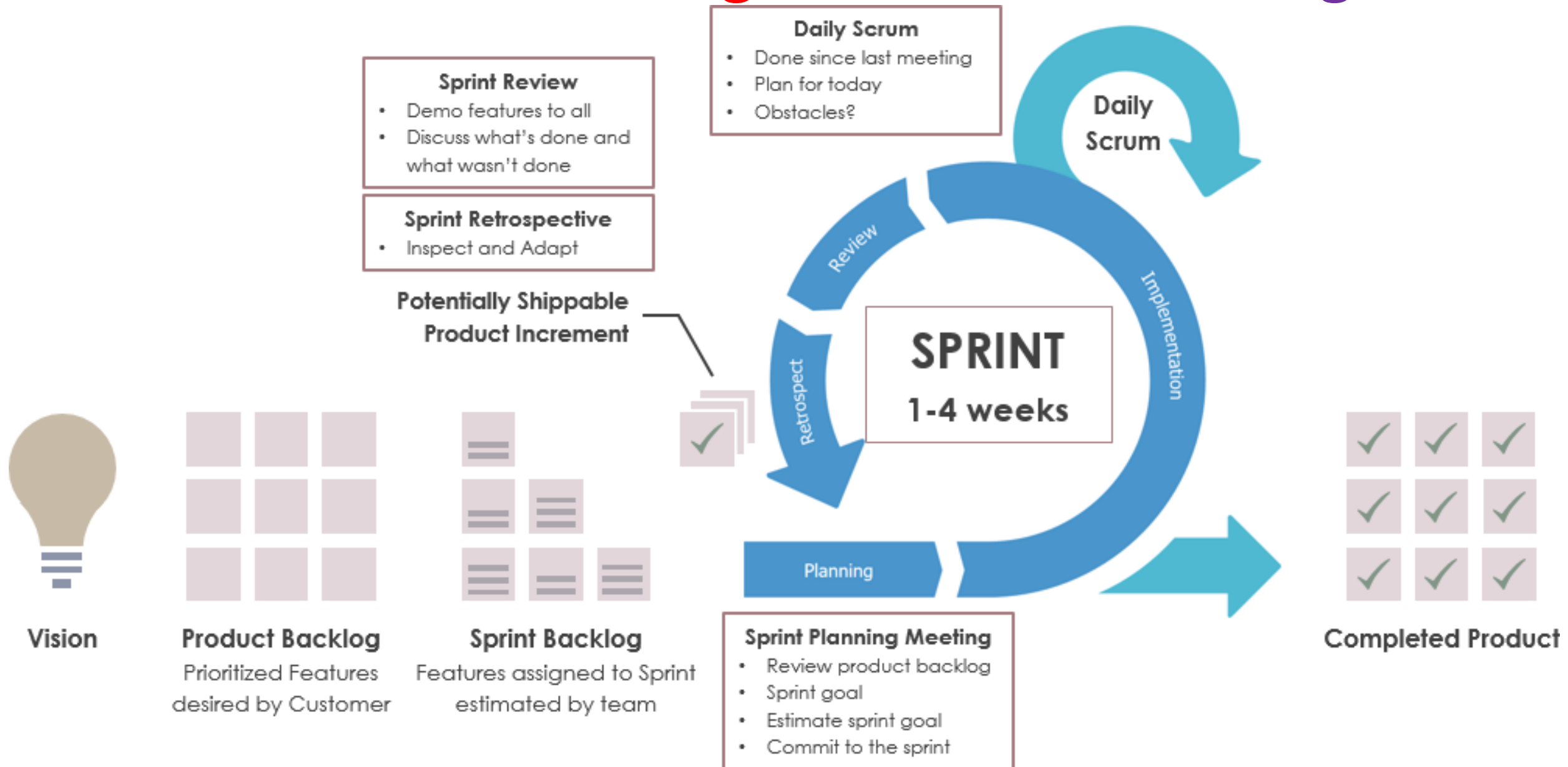
Scrum Terminologies: Sprint Review



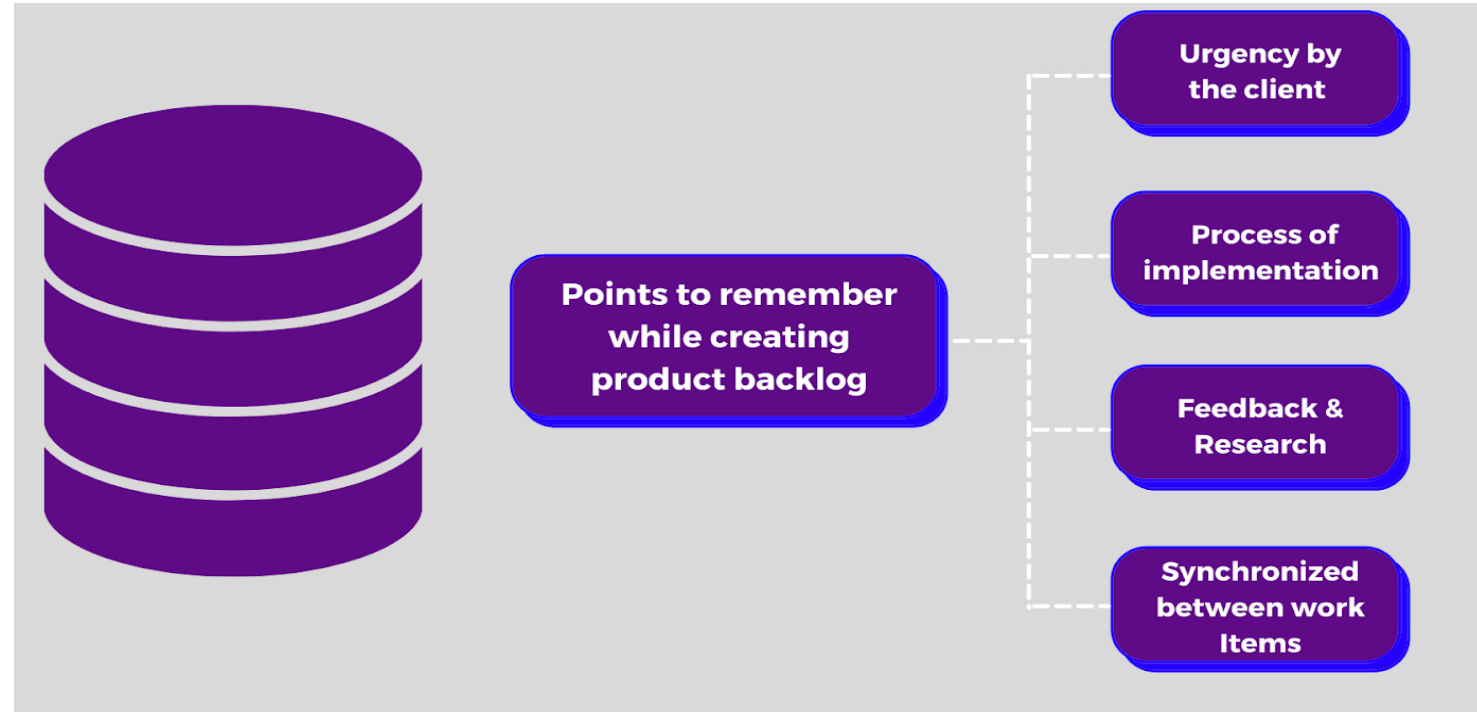
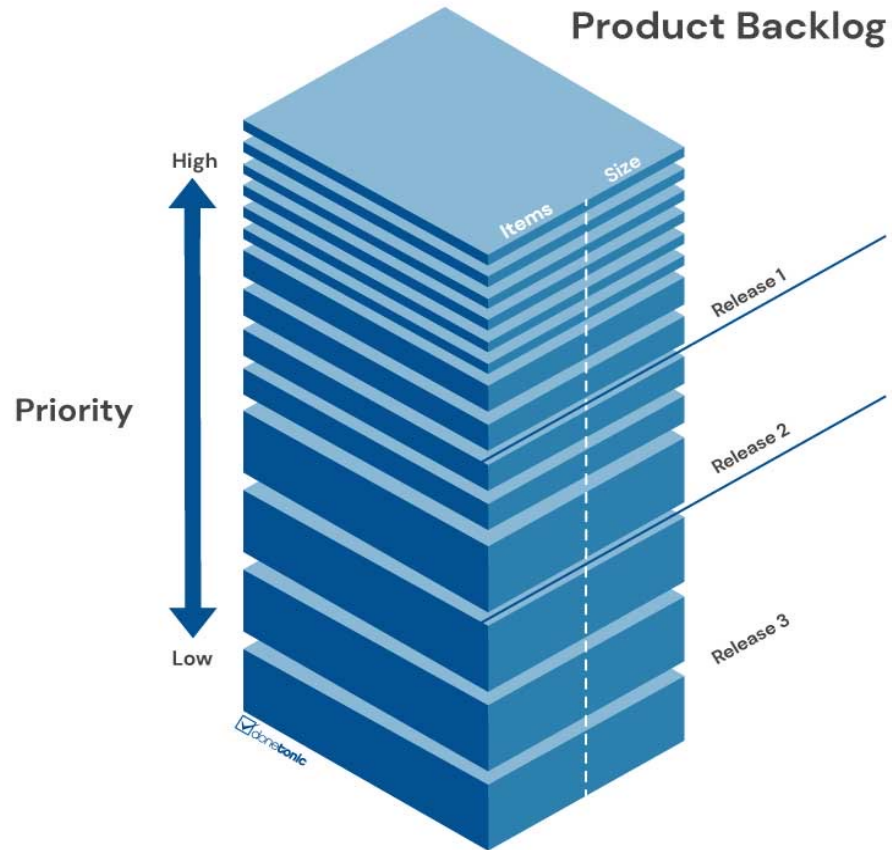
Scrum Terminologies: Retrospective



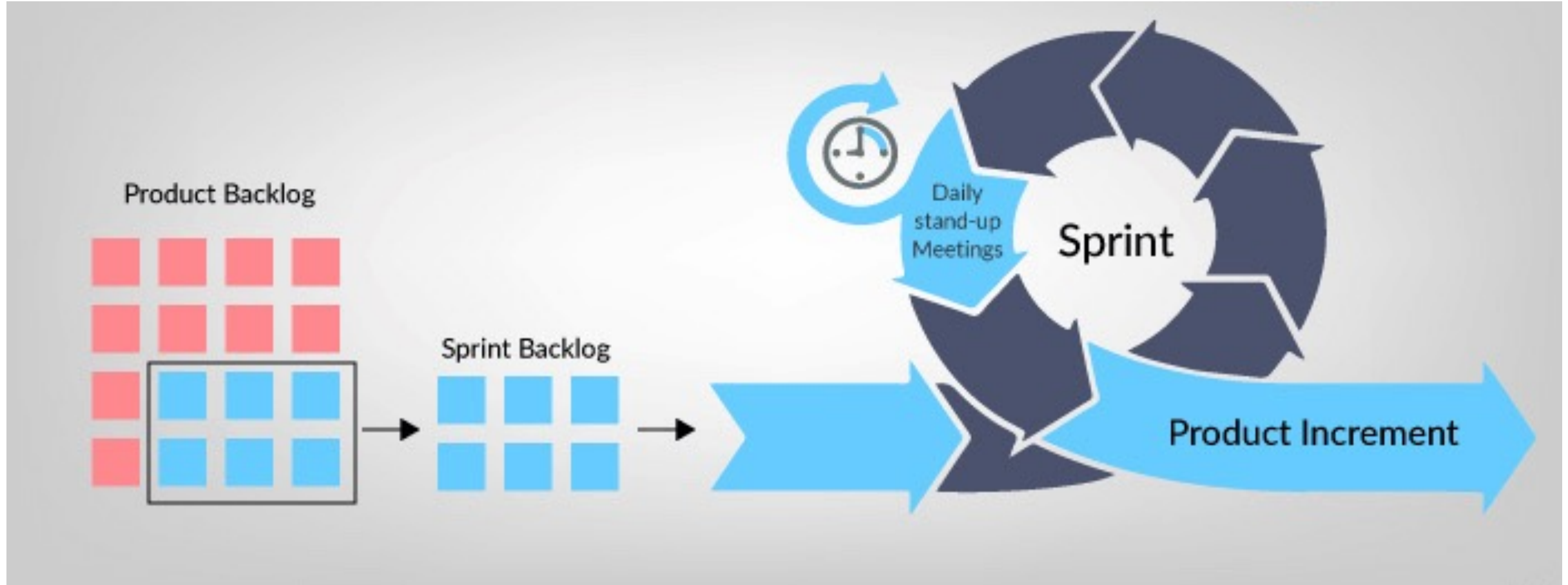
Scrum Terminologies: Product Backlog



Scrum Terminologies: Product Backlog



Scrum Terminologies: Sprint Backlog



Scrum Terminologies: Sprint Backlog

Sprint Backlog

User Stories

Tasks



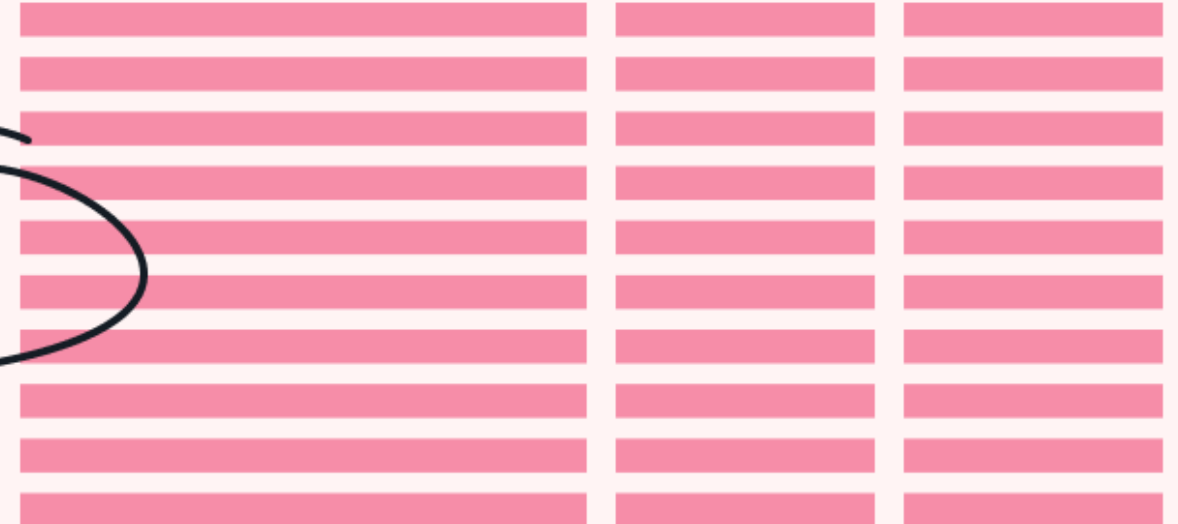
VS

Product Backlog

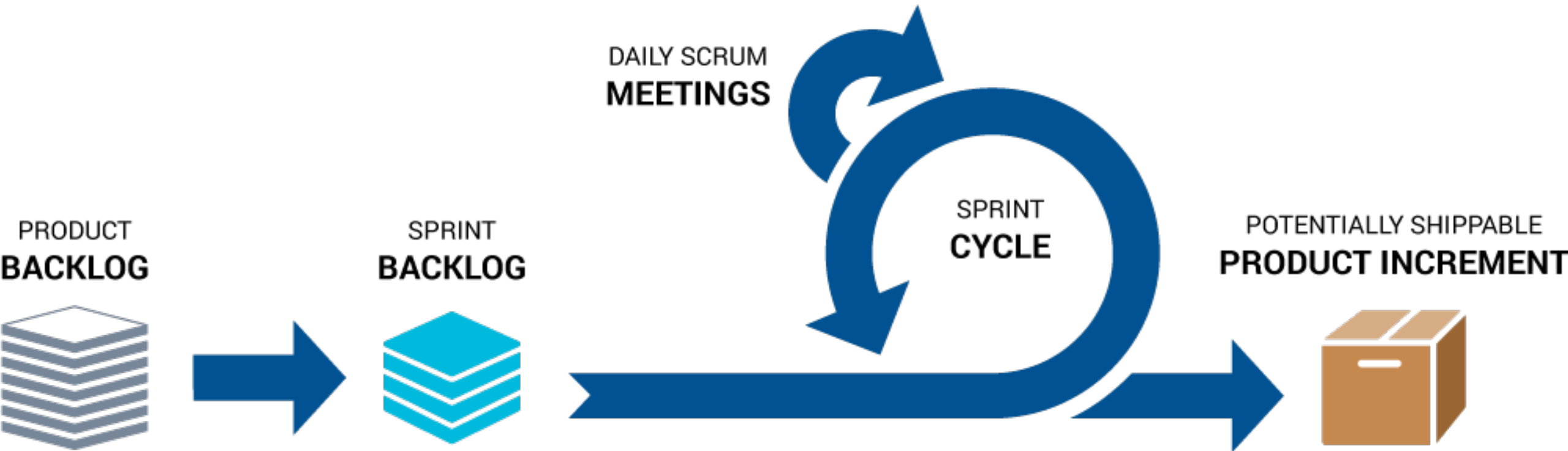
User Stories

Priority

Story Points



Scrum Terminologies: Increment



Scrum Terminologies: Increment



Sprint Goal and Backlog



Sprinting



Done Increment

Best Practices of Scrum



Benefits of Scrum

Time to Market



Increased Transparency



Cost-Effectiveness



Adaptability & Flexibility

